0.18um FlashIP™ Application Note

Version 1.4

Date: July 29, 2003

Description

The FlashIP family have two types of operation modes, the user mode and test mode. The internal charge pump and voltage regulator are built-in for the convenience of erasing or programming operation to the FlashIP with a single 1.8V power supply.

In general, the flash array is divided into two array blocks. One is the main block, and the other is the information block with one only page of flash array. The information block can be used to store the information of the flash devices. A page is defined as a flash memory sector made up of eight adjacent word lines, regardless of the FlashIP configurations. That is, the page size in bits is fixed as eight times the number of bit lines per word line.

A page erase cycle erases one page of flash array to state “1”, and the mass erase cycle erases the whole flash array to state “1”. A program cycle writes a word or words of data bits from the data input bus to the addressed location or locations on the selected word line. No words order is required for the programming operation. No internal data latches are added before or after the FlashIP interface pins, including the data output pins. That is, the data input and data output of the

Description (continued)

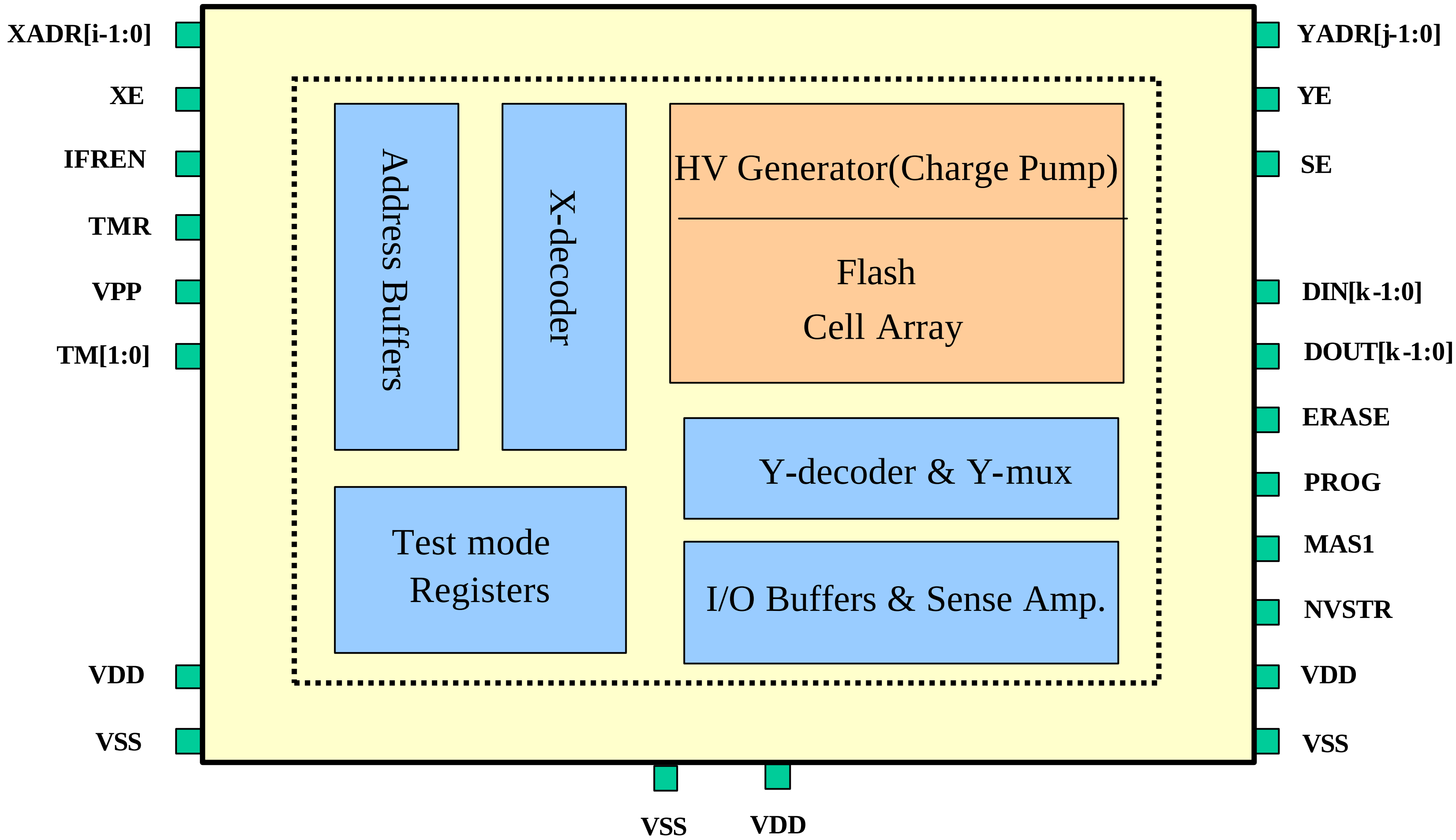
FlashIP family are unlatched or unregistered.

Before the program operation, the erase operation must be done first either by the page erase or mass erase to initialize the selected page or the whole flash array to state "1". A program "0" operation writes the selected flash bit-cells to state "0". However, a program "1" operation is practically no operation to the selected bit-cells, keeping them remained in the initial state of "1". The same addressed location can't be programmed more than twice before next erase.

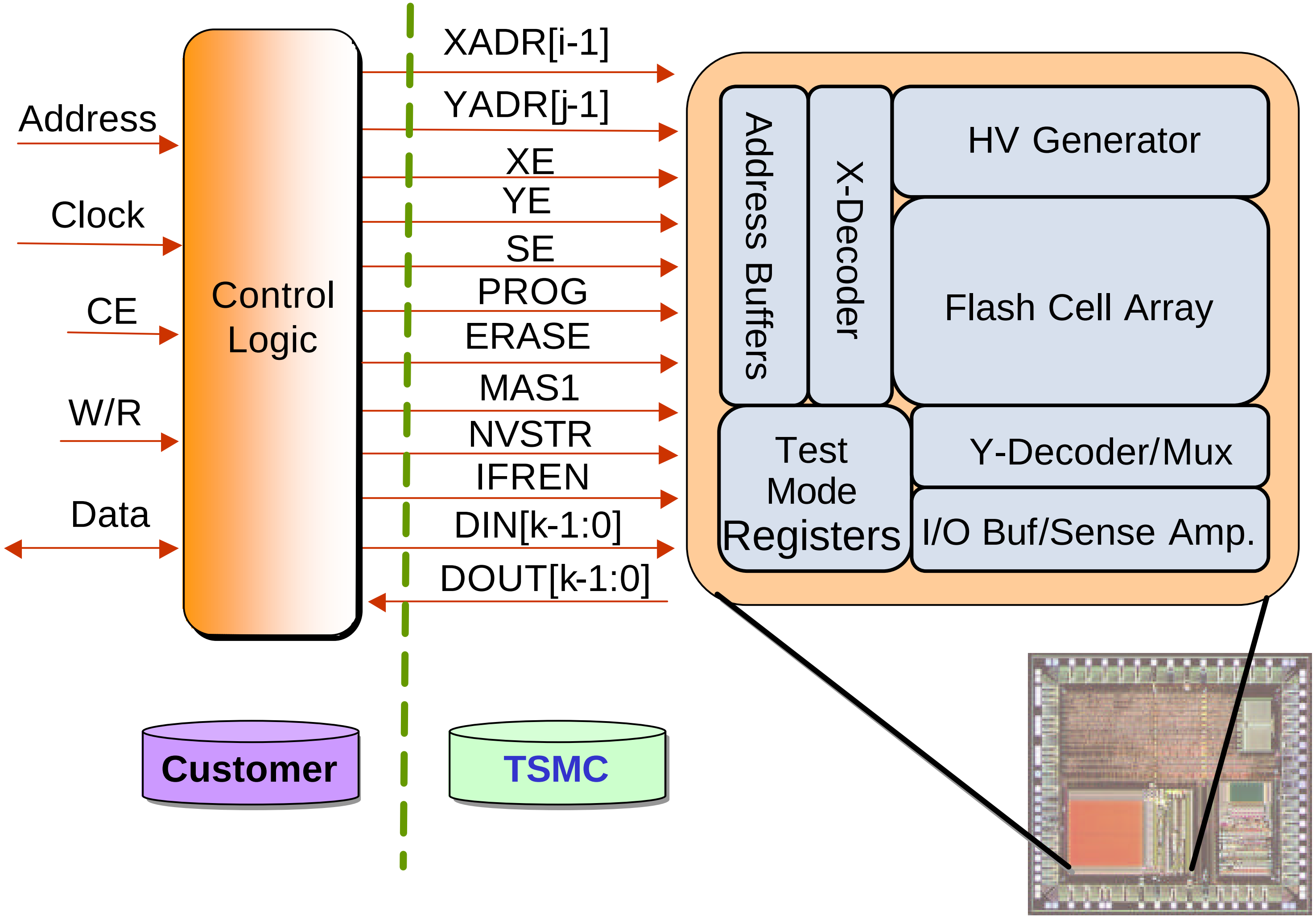
To make the embedded FlashIP blocks testable during the wafer sorting for testing the embedded FlashIP blocks with the proprietary EmbFlash memory test patterns to insure the quality and reliability, all the FlashIP interface pins have to be accessible, controllable and observable, through user I/O pins.

The Design-for-Testability circuitry has to be added for the embedded FlashIP blocks. The Parallel Access Multiplexing methodology among all the DFT approaches is most preferred, providing a direct and simple way for sorting the FlashIP blocks during the wafer level test. This also helps reduce the effort of test program development and the test time needed for the FlashIP blocks.

FlashIP™ Block Diagram & Connectors



FlashIP™ Interface Diagram



FlashIP™ Connector Description

FlashIP User Mode

Connector name	I/O	Description
XADR[i-1:0]	I	X address input, selecting a word line. XADR[i-1:3] select one page from the main block. XADR[2:0] select one of the eight word lines on the selected page of the main block or information block.
YADR[j-1:0]	I	Y address input, addressing one data word on the selected word line.
DIN[k-1:0]	I	Data input bus.
DOUT[k-1:0]	O	Data output bus.
XE	I	X address enable, disabling all the word lines when XE=0.
YE	I	Y address enable, disabling all the Y-mux switches when YE=0.
SE	I	Sense amplifier enable.
IFREN	I	Information block enable, accessing information block when IFREN=1.
ERASE	I	Define the erase cycle for page erase or mass erase.
MAS1	I	Define the mass erase cycle, erasing the whole flash array.
PROG	I	Define the program cycle.
NVSTR	I	Define the non-volatile store cycle, enabling the internal high voltage.

Notes:

1. Need to multiplex all control, address and data signals to I/O pads or package pins for flashIP testing.
2. If the interconnect width of the power line or ground line is as wide as 30um, only connecting one set of VDD/VSS ports is allowable. If not, the total interconnect width connecting to the two or three sets of VDD/VSS ports should be at least 30um wide.
3. The XADR[i-1:3] are don’t care when accessing the information block.

FlashIP™ Connector Descript. (continued)

FlashIP Test Mode

Connector name	I/O	Description
TMR	I	Reset for test mode latches, operating in 0~Vdd. In user mode, it should be always connected to Vdd.
VPP	I/O	A direct high voltage input/output pin, operating in 0~15V. It is mainly used to force current for the mass program test mode in the CP test flow. It can also be used to force high voltage or measure high voltage in other test modes. This pin is defaulted to tri-state in user mode.
TM[1:0]	I/O	Direct input/output pins, operating in 0~Vdd. They can be used to force voltage or current, or measure voltage or current in test mode. These direct pins are defaulted to tri-state in user mode.
Vdd	I	Power supply.
Vss	I	Ground.

Notes:

1. Need to multiplex all the four FlashIP test-related signals to user I/O pads for wafer-level testing when analyzing or debugging the FlashIP, if necessary.
2. The four test pads are not necessarily bonded to package pins.
3. If the capability of faliure analysis or debugging is going to be kept for package parts, four extra package pins should be assigned to the four FlashIP test-related signals.

FlashIP™ Timing Parameter

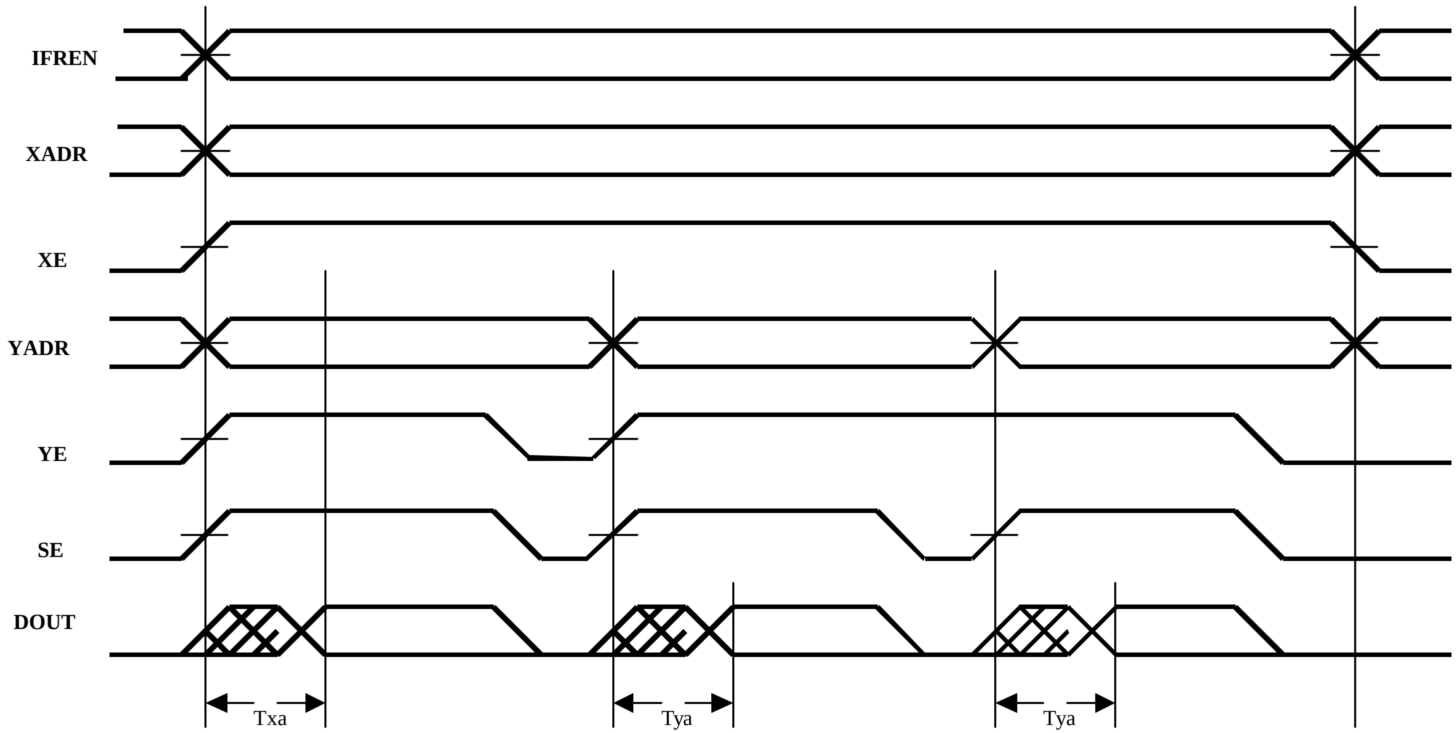
FlashIP User Mode Timing Parameters

Parameter	Symbol	SFD0512_16BC		Unit
		Min.	Max.	
X address access time	Txa	-	50	ns
Y address access time	Tya	-	50	ns
PROG/ERASE to NVSTR set up time	Tnvs	5	-	us
NVSTR hold time	Tnvh	5	-	us
NVSTR hold time (mass erase)	Tnvh1	100	-	us
NVSTR to program set up time	Tpgs	10	-	us
Program hold time	Tpgh	10	-	ns
Program time	Tprog	20	40	us
Address/data set up time	Tads	20	-	ns
Address/data hold time	Tadh	20	-	ns
Recovery time	Trcv	1	-	us
Cumulative program HV period[2]	Thv	-	31	ms
Erase time	Terase	20	-	ms
Mass erase time	Tme	200	-	ms

Notes:
1. All the timing figures are pre-simulated and subject to change before silicon verification.
2. Thv is the cumulative high voltage programming time to the same row before next erase, and the same location can't be programmed more than twice before next erase.
3. All the signals that align together in the timing diagrams should be driven from the same clock edge. Setup and hold times are not critical if they are within 10ns.
4. The reading operation is combinatorial. Sequence of timing edges are not important. The latest valid signal determines the access time.
5. All the timing measurements are from the 50% of the input to 50% of the output.
6. All the input waveforms have rising time(tr) and falling time(tf) of 1ns.
7. For the capacitive loads greater than 1pF, the access time will increase by 1ns per pF of additional loading.
8. The timing figures may be different among the FlashIP blocks. Please refer to the datasheet for the timings of a specific FlashIP block.

Word (Byte) Read

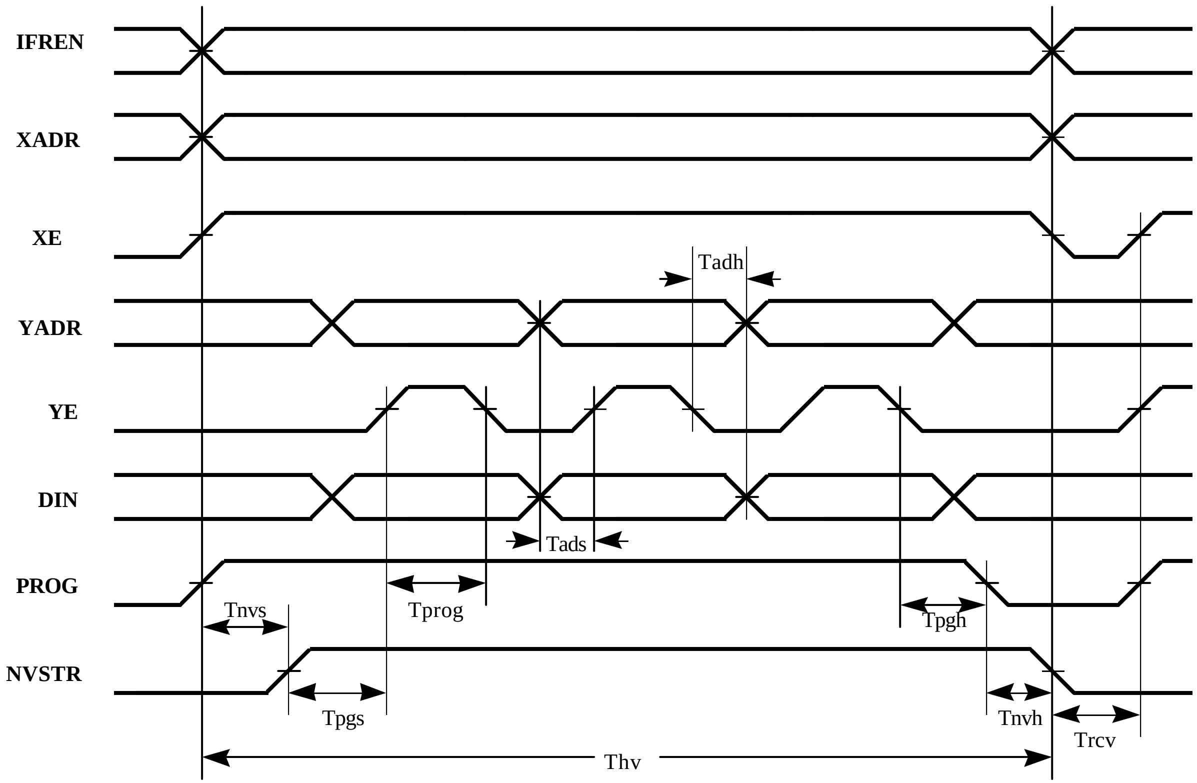
The three control signals XE, YE and SE must be activated. The IFREN control signal is used to enable the access to the information block. The sequence of timing edges is not important, and the latest valid signal determines the access time. Txa is the access time of XADR or XE. Tya is the access time of YADR, YE, or SE. No internal data latches are added for the DOUT.



FlashIP™ User Mode (continued)

● Word (Byte) Program

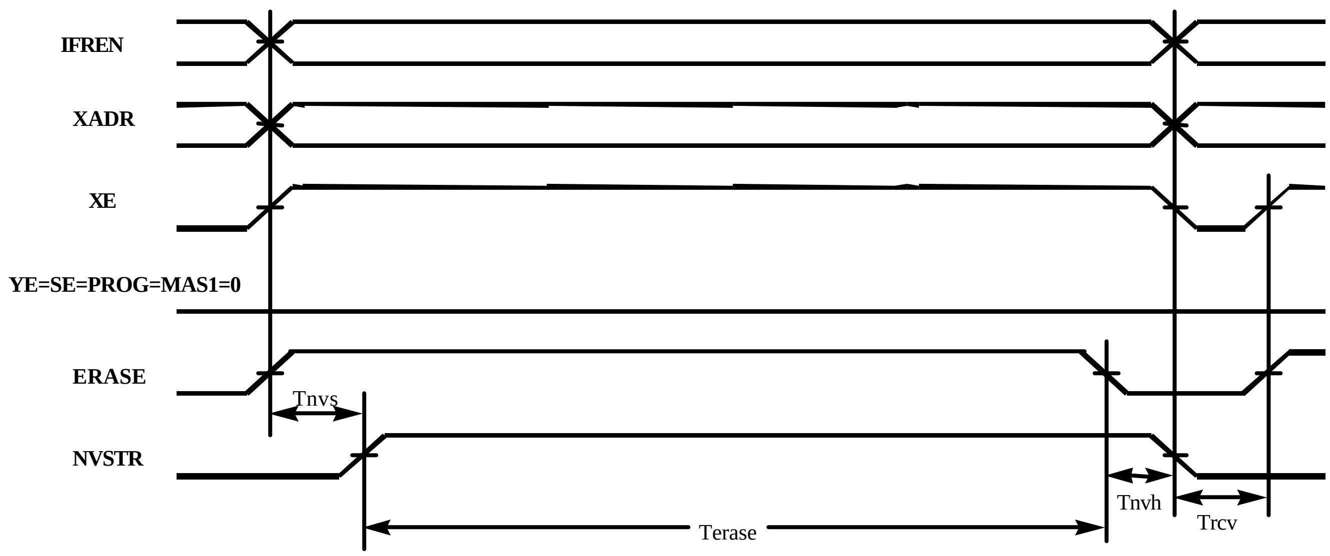
The four control signals XE, YE, PROG, and NVSTR must be activated. The IFREN control signal is used to enable the access to the information block. A multi-word programming within the same program cycle is allowed. But the same location can't be programmed more than twice before next erase.



FlashIP™ User Mode (continued)

● Page Erase

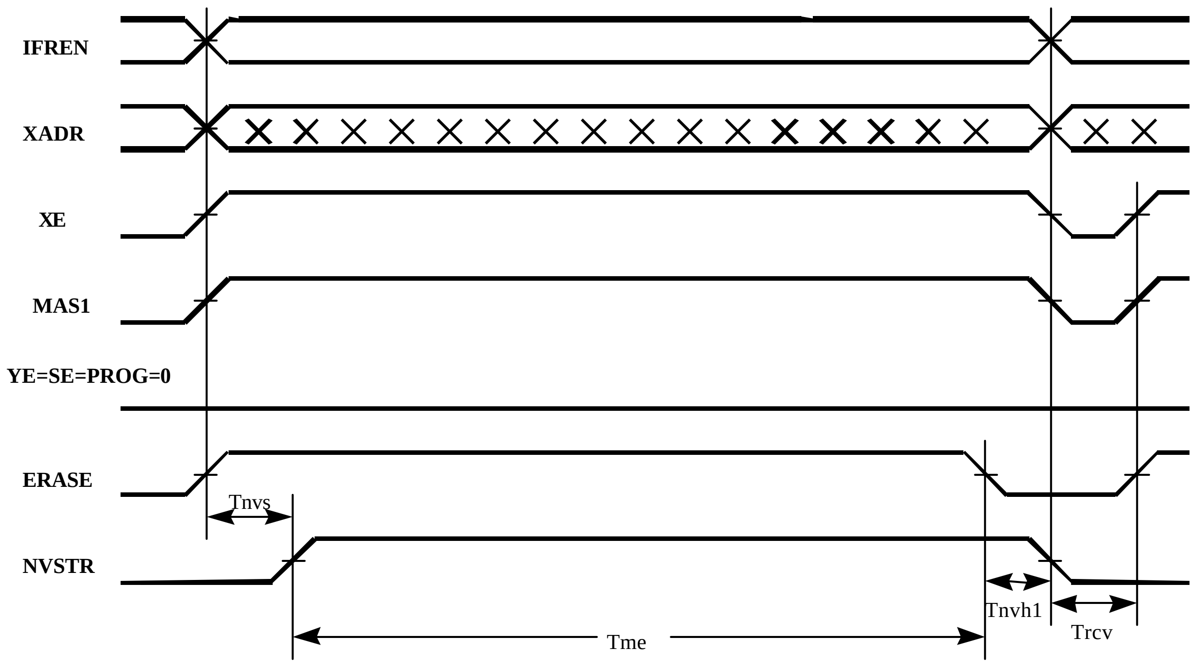
The three control signals XE, ERASE, and NVSTR must be activated, and other control signals must be inactivated. The IFREN control signal is used to enable the access to the information block. The selected page of flash array will be erased to state “1”. The page to be erased is selected by the XADR[i-1:3]. The XADR[2:0] and YADR[n-1:0] are don’t-care during the page erase operation.



FlashIP™ User Mode (continued)

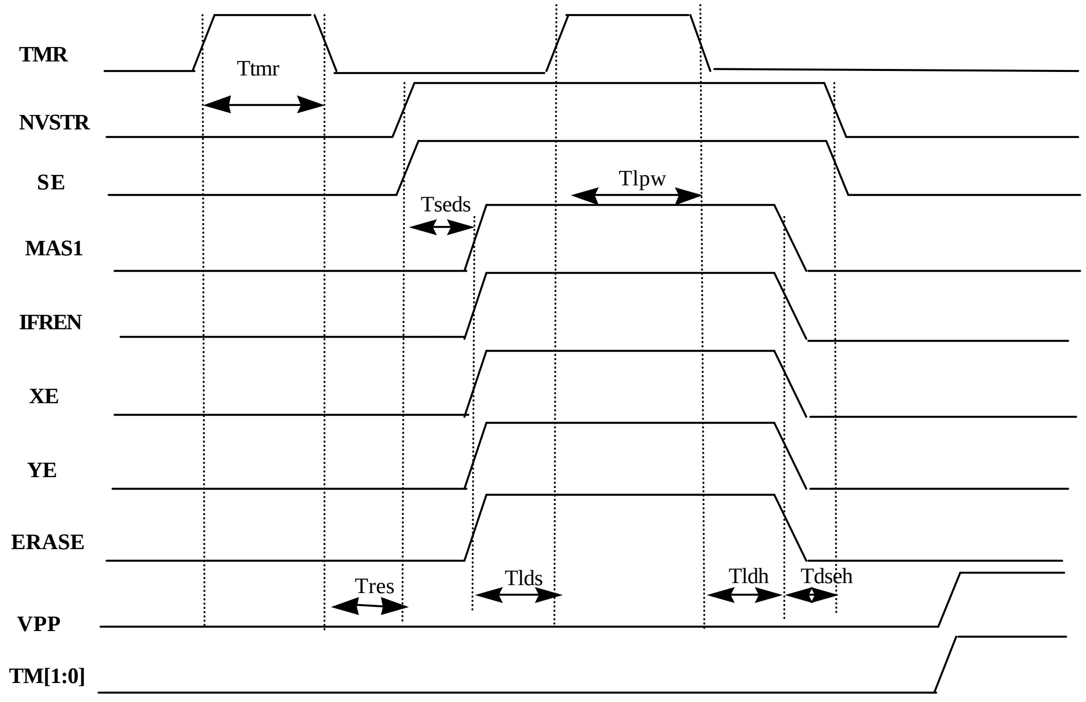
● Mass Erase

The four control signals XE, ERASE, MAS1, and NVSTR must be activated and other control signals must be inactivated. Both the main block and information block are erased to state “1” when the IFREN control signal is enabled. Otherwise, only the main block is erased. All the XADR and YADR are don't-care during the mass erase operation.



FlashIP™ Test Mode

- Three steps to handle the FlashIP test mode:
  - (a) Set TMR to "1" to reset the test mode latches when NVSTR = SE = 0
  - (b) Setting the test mode registers via ERASE, YE, XE, IFREN, MAS1, when TMR = NVSTR = SE = 1
  - (c) Test the FlashIP block with the associated timing diagram of the test mode
- Do not insert any delay circuit or logic circuitry before or after the following analog pins: VPP, TM[1:0].



FlashIP™ Test Mode (continued)

Test Mode Decoder Truth Table

Mode	CODE				
	ERASE	YE	XE	IFREN	MAS1
Test mode 1	L	H	L	L	H
Test mode 2	L	H	L	H	L
Test mode 3	H	H	H	L	L
Test mode 4	H	H	H	L	H
Test mode 5	H	H	H	H	L

Test mode Timing Parameters

Parameter	Symbol	Value		Unit
		Min.	Max.	
TMR pulse width	Ttmr	20	-	ns
TMR to internal TMSE set up time	Trses	10	-	ns
Internal TMSE to data set up time	Tseds	10	-	ns
Data set up time for latch	Tlds	10	-	ns
Set latch pulse width	Tlpw	20	-	ns
Data hold time for latch	Tldh	10	-	ns
Data to internal TMSE hold time	Tdseh	10	-	ns

Note: The timing figures may be different among the FlashIP blocks. Please refer to the datasheet for the timings of a specific FlashIP block.

High Voltage I/O PAD (HV PAD)

This is a direct high voltage input/output pad with ESD protection devices, operating in 0~14V. Though this test pad is not necessarily bonded to a package pin, it is a must for the CP test flow of FlashIP for volume production.

It is specially designed to be solely used along with TSMC's embedded FlashIP blocks. The package pin of the HV pad for other purpose or user application is strongly discouraged.

It is mainly used to force current into the embedded FlashIP blocks for the mass program test mode in the CP test flow to save the test time for setting the whole flash array to background "0". It can also be used to force high voltage bias into the FlashIP or to measure high voltage bias from the FlashIP in other test modes, while analyzing or debugging the embedded FlashIP blocks.

Do not insert any delay or logic circuitry before or after the HV pad. Some shielding techniques may be added for the VPP interconnect outside the embedded FlashIP blocks to avoid the unexpected coupling to or from other sensitive signals of customer circuitry.

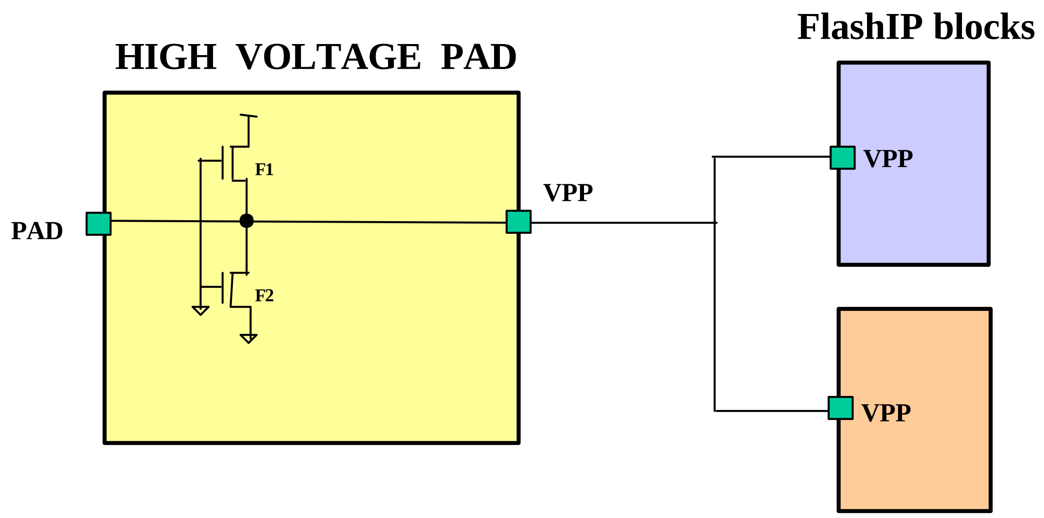
High Voltage I/O PAD (continued)

The VPP pins of the FlashIP blocks are defaulted to tri-state in user mode. By directly wiring the VPP pins of multiple FlashIP blocks to the same HV pad, this test pad can be shared among the FlashIP blocks, as long as the total current needed for the mass program operation does not exceed the maximum current allowable to be forced into the HV pad, 62.36mA@110C.

The current needed for the mass program operation is actually proportional to the number of I/O data bits, 400uA per I/O bit. For example, a FlashIP block with 8 I/Os, 16 I/Os, or 32 I/Os will take 3.2mA, 6.4mA or 12.8mA for the mass program operation.

The Jmax specifications of TSMC 0.18um metal rules @110C are 1mA/um for metals 1-5, 1.6mA/um for metal-6 or top-metal, 0.53mA/contact for Contact, 0.28mA/via for Vias 1-4, 0.706mA/via for Via-5 or top-via, and 0.28mA/via for all the stack Contacts or Vias. The rating factor of the Jmax from 110C to 70C is 3.44.

High Voltage I/O PAD (continued)



NOTE:

1. F1 and F2 are HV NMOS transistors for ESD protection
2. F1 and F2 are symbolic representation of the ESD devices
3. The VPP pin of the FlashIP blocks are defaulted to tri-state

Analog I/O Pads of TM[1:0]

The analog I/O pads for the TM[1:0] pins of the embedded FlashIP blocks are direct input/output pads with ESD protection devices, operating in 0~Vdd. These analog I/O pads are served as two FlashIP specific test pads for the TM[1:0] pins. They are not necessary to be bonded to packaged pins.

The two pads can be used to force voltage or current into the embedded FlashIP blocks, or to measure voltage or current from the FlashIP in test mode, while analyzing or debugging the FlashIP blocks. The maximum current allowable to be forced into the TM[1:0] pins or measured from the TM[1:0] pins is 1mA.

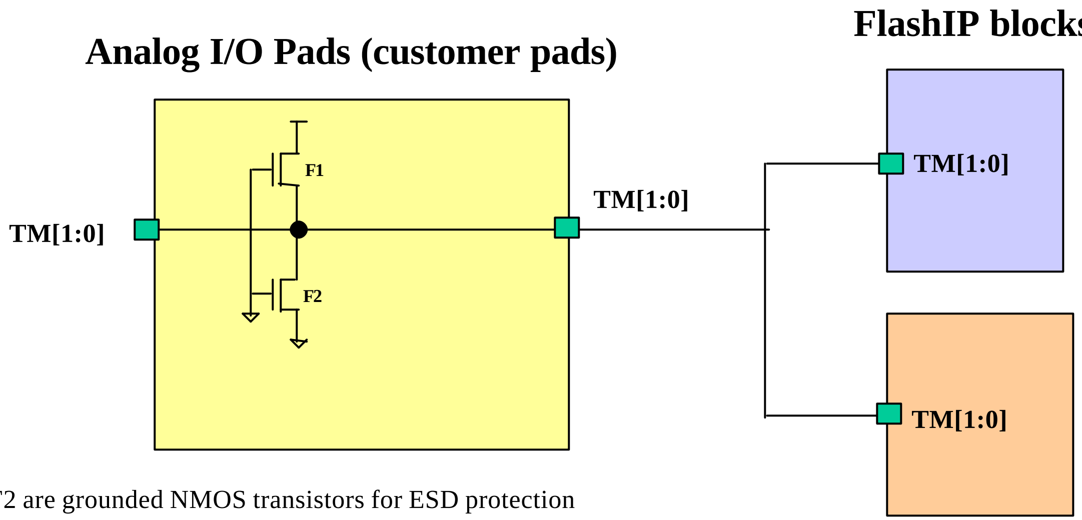
Do not insert any delay circuit or logic circuitry before or after the TM[1:0] test pads. Some shielding techniques may be added for the interconnects of the TM[1:0] pins outside the embedded FlashIP blocks to avoid the unexpected coupling to or from other sensitive signals of customer circuitry.

The TM[1:0] pins of the FlashIP blocks are defaulted to tri-state in user mode. By directly wiring the TM[1:0] pins of multiple FlashIP blocks to the same analog TM[1:0] pads respectively, these two test pads can be shared among the multiple FlashIP blocks.

Analog I/O Pads of TM[1:0] (continued)

The analog I/O cell of PDIANA2 from the DSD/TSMC 0.18um analog I/O library for low frequency application can be used as one of the alternatives for the analog I/O pads of TM[1:0]. Or customers can use their own direct I/O pads for the test pins of the TM[1:0] for the embedded FlashIP blocks.

The Jmax specifications of TSMC 0.18um metal rules @110C are 1mA/um for metals 1-5, 1.6mA/um for metal-6 or top-metal, 0.53mA/contact for Contact, 0.28mA/via for Vias 1-4, 0.706mA/via for Via-5 or top-via, and 0.28mA/via for all the stack Contacts or Vias. The rating factor of the Jmax from 110C to 70C is 3.44.



NOTE:

1. F1 and F2 are grounded NMOS transistors for ESD protection
2. F1 and F2 are symbolic representation of the ESD devices
3. The TM[1:0] pins of the FlashIP blocks are defaulted to tri-state

Design Considerations

- **No hardware protection circuitry or software algorithm is built-in for the FlashIP blocks to prevent inadvertent or unwanted erase or program operation to the FlashIP memory array.**

  So the FlashIP users have to add some kind of hardware protection circuitry or software algorithm outside the embedded FlashIP blocks to avoid the accidental erase or program operation to the FlashIP, especially during the power-on or power-off transient.

- Specifically, concerning the the power-on transient of the power supply, all the control signals of the FlashIP blocks should be gated off by the on-chip power-on reset or the external reset driven from other power-on reset. For example, the XE, YE, SE, PROG, ERASE, MAS1, NVSTR, IFREN are gated off to “0”, and the TMR is gated off to “1” during the power-on period.

- If the power-off is also posed as a concern for the possible undesired erase or program operation to the embedded FlashIP blocks, the hardware or software protection scheme for the power-off transient should also be implemented.

- There is no redundancy circuitry implemented for the FlashIP blocks.

Design Considerations (continued)

- The TMR pad, HV pad and the two TM[1:0] analog pads are the four FlashIP test-specific pads needed for the wafer level sorting, analysis, or debugging. They are not necessarily be bonded to package pins.

- The TMR pad and HV pad are must in the CP test flow of FlashIP for volume production. The TMR pad along with the multiplexed NVSTR, SE, MAS1, IFREN, XE, YE and ERASE, is used for setting the mass program test mode and other test modes. The HV pad is mainly used for forcing current into the FlashIP blocks for the mass program operation.

- The two TM[1:0] analog pads are must to preserve the capability of failure analysis or debugging to the embedded FlashIP blocks at the wafer level stage for the possible returned failure of EmbFlash products, once when necessary.

- The internal charge pump and voltage regulator are built-in for the convenience of erasing or programming operation to the FlashIP blocks with a single 1.8V power supply.

- In general, the flash array is divided into two array blocks. One is the main block, and the other is the information block with one only page of flash array.

Design Considerations (continued)

The information block can be used to store the information of the flash devices.

- A page is defined as a flash memory sector made up of eight adjacent word lines, regardless of the FlashIP configurations. That is, the page size, in bits, of a FlashIP block is fixed as eight times the number of bit lines per word line.

- The number of word lines per page for the EE emulators may be smaller than eight to minimize the number of words per page, for example, 4 words/page or 8 words/page.

- All the control signals of FlashIP blocks must be generated by the user control or glue logic. They are preferably driven from the same system clock edge. Please refer to the timing waveforms of FlashIP datasheets.

- In the program cycle, the FlashIP data may be destroyed accidentally if the timings and waveforms for program are not followed. For example, the change of the XADR, when XE = 1, or the change of the YADR, when YE = 1, during the program cycle.

- In the page erase cycle, the FlashIP data may be destroyed accidentally if the

Design Considerations (continued)

timings and waveforms for page erase are not followed. For example, the change of the XADR[i-1:3], when XE = 1, during the page erase cycle.

- No recovery time is required from standby to the first word (byte) read operation or the first word (byte) program operation.
- The access time of SE is equal to Tya, Y-address access time.
- The X-address is used to select one of the word-lines. The Y-address is used to select a word of bit-lines on the same selected word line.
- No setup time or hold time is required between the XADR and XE for the read, program, or erase cycle.
- In a read cycle, the minimum pulse width of the YE enable in Logic Low is 0 ns. In a program cycle, the minimum pulse width of the YE enable in Logic Low of is equal to "Tads + Tadh".
- The XE is not required to be toggled when the XADR change to a new X-address in a read cycle.

Design Considerations (continued)

- A high-to-low transition of XE, thai is, inactivating the XE, is required when the XADR change to new X-address in a program cycle.

- The read current, Idd, of a FlashIP block with cycle time T can be calculated by the following equation:

  Idd = (Idd1 - Idd2) x Tmin / T + Idd2

  Where

  - Idd1 is the read current when T equals Tmin
  - Idd2 is static read current
  - Tmin equals Txa, the minimum allowable cycle time
  - T is the cycle time

- The FlashIP will consume more power when SE is active. Keeping SE inactive for non-read cycles can reduce some power drawn from sense amplifiers.

- Users can append 3-state buffers after DOUT, if the 3-state scheme is to be used for internal data bus of the EmbFlash product.

Design Considerations (continued)

- The internal 3-state bus is discouraged from the viewpoints of the static timing analysis, scan insertion, power consumption, bus contention, and even the bus performance.

- All the input waveforms of the FlashIP blocks have the rising time (tr) and falling time (tf) of 1ns. For the capacitive loads greater than 1pF, the access time will increase 1ns/pF. That is, the driving capability of the DOUT is 1ns/pF for additional capacitive loads. External output buffers are needed to be added after the DOUT to drive large capacitive loads.

- It is not allowed to do read operation during the erase or program cycle.

- In the mass erase cycle, all the flash cells are erased to "1". In the page erase cycle, only the flash cells of selected page are erased to "1".

- When the ERASE, XE, and NVSTR are all set high, the selected page of flash bits will be erased. When the ERASE, XE, MAS1, and NVSTR are all set high, the whole flash memory array will be erased.

- Before a valid program or write operation, the FlashIP must be erased first by the page erase or mass erase to set the initial state of the flash memory to “1”.

Design Considerations (continued)

However, the same memory location can not be programmed more than twice before next erase.

- The probability of a cell being disturbed is proportional to the cumulative program time on the same word line before next erase.

- The cumulative program time of high voltage to the same word line must be smaller than the Thv specification as specified in the datasheet, while the NVSTR is active continually before the the disable of the XE for switching XADR or finish of programming.

- The cumulative program time (Thv) is applicable for the same word line, row, before it is erased again either by the page erase or mass erase.

- In a program cycle, the programming "0" operation means to write data "0" into the selected flash cells, and the programming "1" operation, in fact, means no operation to the selected flash cells, thus keeping them unchanged.

- If a program or erase cycle is interrupted before the completion of the operation, the FlashIP data of selected become unknown, maybe partially programmed or erased.

Design Considerations (continued)

- The program and erase operations to the same FlashIP block should not be asserted at the same time.

- If the multiple FlashIP blocks are to be word-programmed or page-erased concurrently during wafer screening, some control logic must be implemented to halt the word program or page erase operation to the FlashIP blocks whose XADR or YADR reach its maximum boundary earlier than those of others. This extra address decoding helps to get around the over program or over page-erase dilemma for some of the FlashIP blocks during the wafer screening.

- In user mode, the TMR pin of FlashIP should be always connected to VDD to reset the test mode latches before or during the user applications. In fact, the test mode latches of FlashIP blocks are reset under the condition. The TMR is set to "1", and the either the SE , or NVSTR, or both are set to "0".

- The test mode latches are set through the ERASE, YE, XE, IFREN and MAS1 pins of FlashIP blocks, when the TMR, SE and NVSTR pins are all asserted High. The data outputs of the five test mode latches are further decoded for a variety of test modes needed for the analysis and debugging of embedded FlashIP blocks at wafer level or at package level.

Layout Considerations

- **All the multiple FlashIP blocks embedded on the same chip are only allowed to be placed in the same orientation, either horizontally or vertically. And the mirror of FlashIP blocks to the X-axis or Y-axis is also allowable. Otherwise, the yield may be unstable, and the process window of yield improvement may be narrower than that of those placed in the same orientation.**

It is strongly recommended to place all the FlashIP blocks horizontally, in 0 degree or 180 degrees. If not, customers should remind TSMC the vertical orientation of the placed FlashIP blocks on the Mask Tooling Form.

- **It is extremely important for the customer layout database to follow the TSMC standard layer definition of the 0.18um processes. Otherwise, the layout database will not be accepted for FlashIP merge before the product tape-out. The layer definition description file could be downloaded from the TSMC-ONLINE web-site.**

Furthermore, if there exists conflict in layer definitions between the customer layout and FlashIP layout databases, it would cause the FlashIP merge process to be error-prone and time-consuming.

Layout Considerations (continued)

- The precision and step size of the layout grid for the FlashIP layout database are 0.001um and 0.005um respectively. All the FlashIP geometry polygons are aligned at this step size, 0.005um.

- The “FLASH” dummy layer, with GDS number 94, is used to mark the boundary of the FlashIP phantoms and HV pad layout. This dummy layer is also required for the logic operations of EmbFlash products on the Mask Tooling Form to distinguish the area of the HV pad and FlashIP blocks from others of customer circuitry.

- To stream-in the GDS phantom of the FlashIP blocks or the GDS layout of the high voltage pad, please double check the layout database to make sure there will be no truncation error due to layout grid discrepance between the FlashIP database and user database.

- The metal-1, metal-2 and metal-3 are used for the FlashIP layout. And some spare grounded shielding in metal-3 layer are built-in over the flash memory array. However, no grounded shielding are added over the peripheral circuit of the FlashIP blocks.

Layout Considerations (continued)

- No routing tracks of metal layers 1-3 are allowable to cross over the flash memory array of FlashIP blocks. And no routing tracks of metal layers 1-4 are allowable to cross over the peripheral circuit of FlashIP blocks.
- The routing tracks of the metal-4 layer are only allowed to be routed over the flash memory array. For routing tracks of metals 5-6 layers, they can be routed freely over the FlashIP blocks.
- The interconnects for the FlashIP interface pins can use the minimum width rules, except for the interconnects of the VPP and TM[1:0] pins.
- The minimum current, forcing into the VPP pad, required for the mass program operation depends on the total number of data bits of all the FlashIP blocks which share the same VPP pad and will be mass-programmed currently.
- The current needed for the mass program operation is actually proportional to the number of I/O data bits, 400uA per I/O bit. For example, a FlashIP block with 8 I/Os, 16 I/Os, or 32 I/Os will take 3.2mA, 6.4mA or 12.8mA for the mass program operation.

Layout Considerations (continued)

- The signal path from the HV pad to the VPP pin of FlashIP should be as short as possible to avoid the potential significant IR drop of the high voltage on the interconnect. Once when this phenomenon surfaces, the high voltage level arrive at the VPP pins of the FlashIP blocks will be lower than expected, causing weak mass program.

- The maximum current allowable to be forced into or measured from the TM[1:0] pins of FlashIP blocks is 1mA.

- The maximum current allowable to be forced into the HV pad, is limited to the VPP port width of the VPP Pad layout, 62.36um wide in metal-2. That is, the Jmax of the VPP port on the VPP pad is limited to 62.36mA@110C.

- Avoid to put other analog blocks in the neighborhood of the embedded FlashIP blocks. If unavoidable, some kind of shielding, like a double guard ring, must be put between the analog blocks and FlashIP blocks.

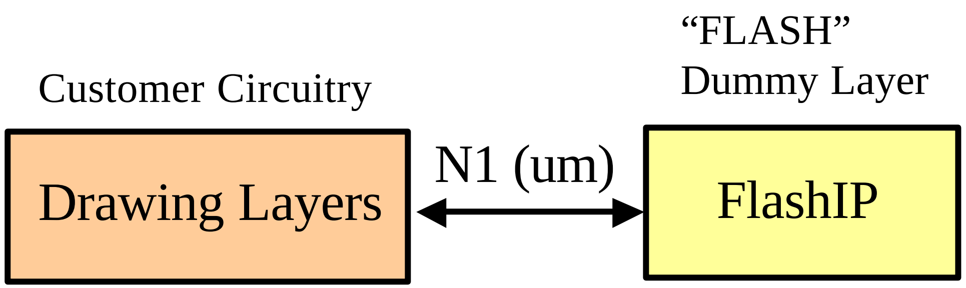
- Make sure all the geometry polygons of the FlashIP phantom are not changed in P&R database. Are the pin widths changed ? Are the pin locations changed ? Has the FlashIP boundary been changed ?

Layout Considerations (continued)

- The FlashIP boundary is marked by the “FLASH” dummy layer, with GDS number 94. The minimum clearance rules from the drawing layers of customer circuitry to the FlashIP boundary to are specified rule by rule as tabulated.

LAYER	N1 (um)
HVNW	4.5
PWELL	4.5
NWELL	4.5
PSUB	1.5
DIFF	2.0
OD2	2.0
HVNT	1.0
POLY1	1.0
N3V	1.0
N2V	1.0
P3V	1.0
P2V	1.0
NP	1.0

LAYER	**N1 (um)**
PP	**1.0**
RPO	**1.0**
ESD	**1.0**
CO	**1.0**
DNW	**5.0**
CTM	**1.5**
HRI	**1.5**
M6T	**3.0**



Multiple FlashIP™ Considerations

Test Time Reduction

The mass program, word program and page erase operations can be performed simultaneously to the multiple FlashIP blocks during the wafer sorting to save the test time needed.

Over Program and Over Page Erase

If the multiple FlashIP blocks are to be word-programmed or page-erased concurrently during wafer screening, some control logic must be implemented to halt the word program or page erase operation to the FlashIP blocks whose XADR or YADR reach its maximum boundary earlier than those of others.

This extra XADR or YADR decoding helps to get around the over program or over page-erase stress to some of the FlashIP blocks with smaller XADR or YADR address during the wafer screening.

Layout Considerations

All the multiple FlashIP blocks embedded on the same chip are only allowed to

Multiple FlashIP™ Consid. (continued)

be placed in the same orientation, either horizontally or vertically. And the mirror of FlashIP blocks to the X-axis or Y-axis is also allowable. Otherwise, the yield may be unstable, and the process window of yield improvement may be narrower than that of those placed in the same orientation.

It is strongly recommended to place all the FlashIP blocks horizontally, in 0 degree or 180 degrees. If not, customers should remind TSMC the vertical orientation of the placed FlashIP blocks on the Mask Tooling Form.

● High Voltage I/O Pad

The VPP pins of the multiple FlashIP blocks are defaulted to tri-state in user mode. By directly wiring the VPP pins of multiple FlashIP blocks to the same HV pad, this test pad can be shared among the FlashIP blocks, as long as the total current needed for the mass program operation does not exceed the maximum current allowable to be forced into the HV pad, 62.36mA@110C.

The current needed for the mass program operation is actually proportional to the total number of I/O data bits, 400uA per I/O bit. For example, the two FlashIP blocks sharing the same VPP pad with 8 I/Os and 16 I/Os respectively

Multiple FlashIP™ Consid. (continued)

will take total 9.6mA for the concurrent mass program operation.

● TM analog pads of TM[1:0]

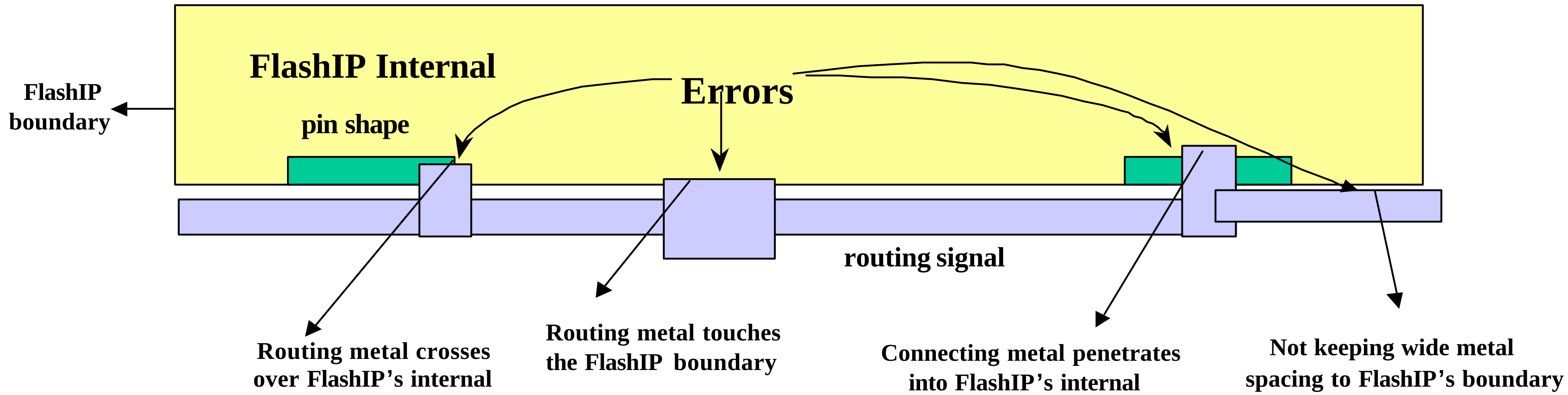
The TM[1:0] pins of the FlashIP blocks are defaulted to tri-state in user mode. By directly wiring the TM[1:0] pins of multiple FlashIP blocks to the same analog TM[1:0] pads respectively, these two test pads can be shared among the multiple FlashIP blocks. The maximum current allowable to be forced into the TM[1:0] pins or measured from the TM[1:0] pins is 1mA.

Most Layout Errors

- The top cell names of the FlashIP blocks are not identical to the original top cell names of the released phantom views.
  - Case sensitive mismatch
  - Customers renamed the top cell names

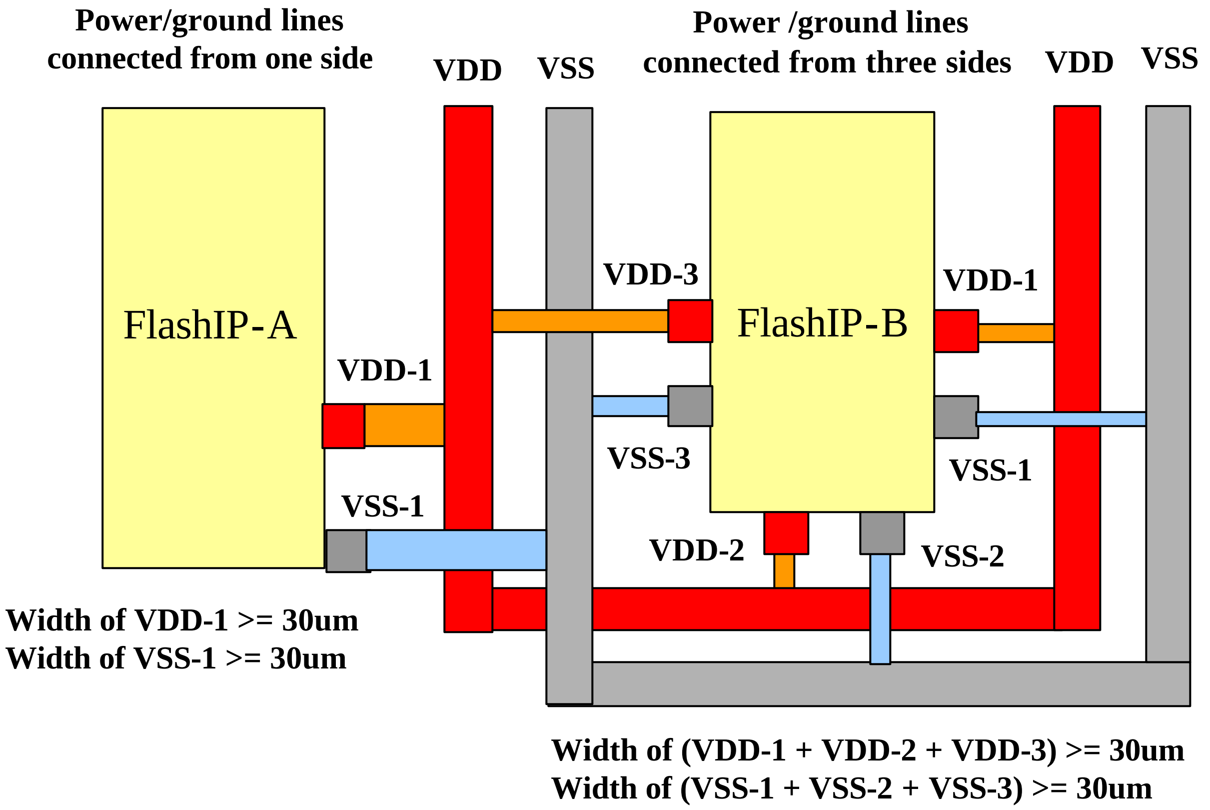
- Shapes of user routing tracks outside the FlashIP blocks overlap the FlashIP area, which does not belong to part of the port area for FlashIP pins.

- Metal pins of customer blocks or cells are not defined by the pin text layers of "40, 41, 42, 43, 44, 45” for metals 1-6 respectively.



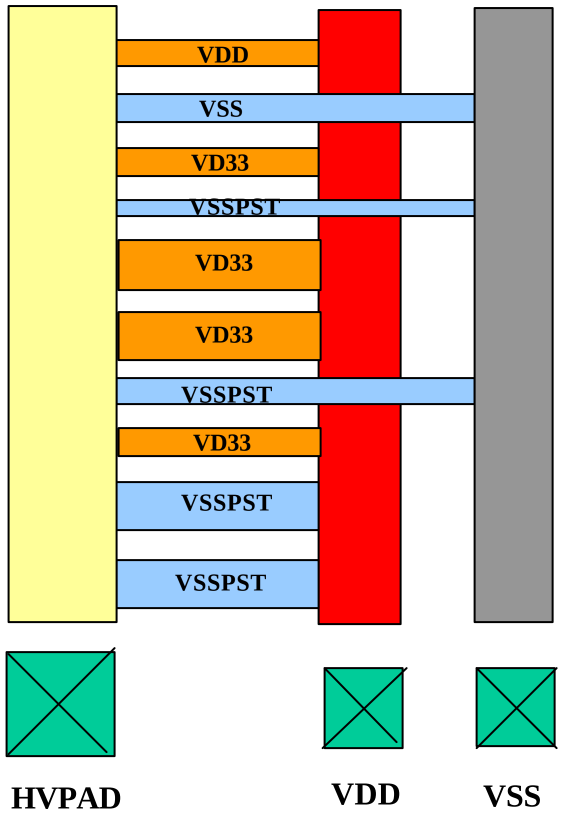
Power Bus Requirement

The FlashIP family provide two or three pairs of VDD/VSS ports for external power and ground lines. Connecting to the external VDD/VSS rings with one only pair, or two pairs, or three pairs of VDD/VSS ports are all allowable, as long as the total widths of the VDD/VSS interconnects are more than 30um wide.

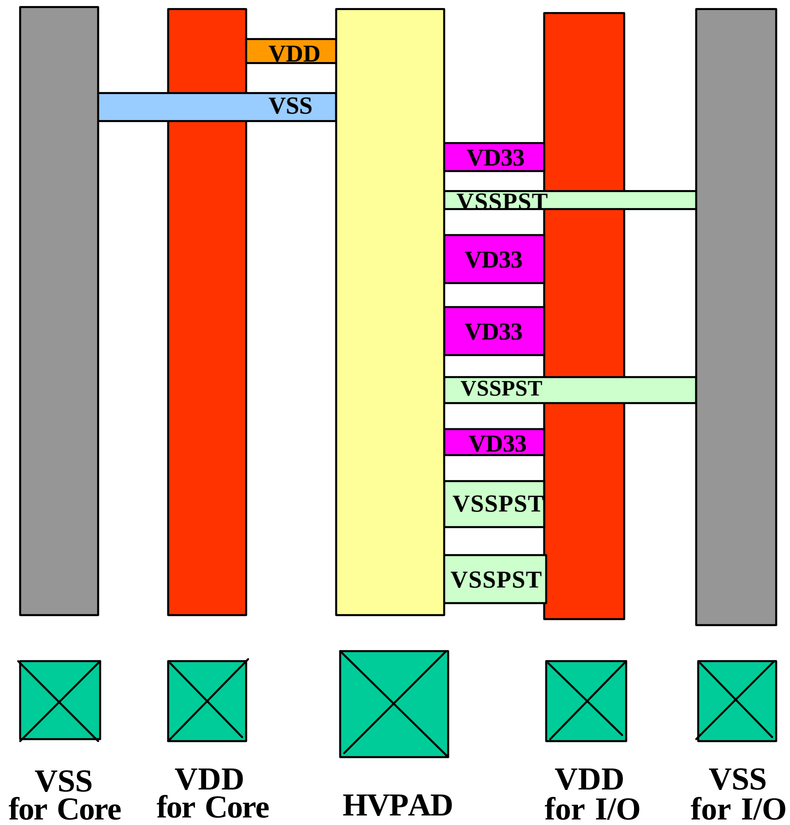


HV PAD VDD/VSS connections

Core circuit and I/O cells share the same power/ground lines



Core circuit and I/O cells use separate power/ground lines



Note : VDD, VSS - Power and ground lines for core circuit
VD33, VSSPST - Power and ground lines for I/O post driver

DRC/LVS Considerations

- **Users should provide the clean DRC/LVS summary reports, customized DRC/LVS run decks if there exist, optional SPICE netlist for LVS check after FlashIP merge, and filled FlashIP Merge Sevice Request Form, along with the GDSII database for the FlashIP merge request.**

- **If the users DRC/LVS run decks are not exactly identical to the original ones provided by TSMC on TSMC-ONLINE, users should provide the customized DRC/LVS run decks to TSMC for the physical verification after FlashIP merge is done. Otherwise, the DRC/LVS checks can not be re-done after FlashIP merge to guarantee the correct operation of the GDSII database merge.**

- The user layout database should follow the TSMC standard layer definition of 0.18um processes. The standard layer definition description file could be downloaded from the TSMC-ONLINE web-site.

- The DRC and LVS verifications and the connectivity among the FlashIP blocks, HV pad and the customer blocks should all be checked successfully by users, before the tape-in of GDSII database to TSMC for FlashIP merge.

DRC/LVS Considerations (continued)

- With the hierarchical LVS check, the connectivity of interface signals between the customer blocks or cells and the FlashIP blocks could be entirely checked. It can be done by the declaration of the FlashIP phantoms as black boxes along with the input of the empty SPICE netlists of the FlashIP blocks.

- Users are recommended to provide the SPICE netlist of the whole chip for FlashIP merge. The correctness of signal connections and electrical rules, open or short, between the FlashIP blocks and user circuitry can then be assured by the LVS check after FlashIP merge.

- The correctness of electrical connections between the FlashIP and user circuitry will not be guaranteed for FlashIP merge, if the user SPICE netlist of the whole chip is not provided to TSMC.

Design for Testability

- All the interface pins of the embedded FlashIP blocks are needed to be accessible, that is, controllable and observable, through user I/O pads or I/O pins for the wafer level screening or for the package parts screening. The TMR, VPP and TM[1:0] test pads are required for the wafer level screening of FlashIP blocks, though the four package pins are optional for package parts.

- If the capability of failure analysis and debugging for the embedded FlashIP blocks are to be kept for package parts, the four TMR, VPP and TM[1:0] pins should be bonded for the Parallel Access Multiplexing and Data/Address Bus Multiplexing approaches. For the Control/Data/Address Multiplexing and Serial Control/Data/Address Multiplexing approaches, Three VPP and TM[1:0] package pins are required. The TMR signal could be generated from the latched data for the data mode of extra signals.

- If the DFT methodology for the embedded FlashIP blocks does not belong to one of the four approaches introduced below, it is necessary to confirm with TSMC to check if the ad-hoc DFT methodology is feasible for the test program development and the testability of the FlashIP blocks.

Design for Testability (continued)

Four multiplexing DFT, Design for Testability, approaches for the embedded FlashIP are introduced here, according the number of user pads or pins, from high to low, available for the test of the FlashIP blocks.

- Parallel Access Multiplexing
- Data/Address Bus Multiplexing
- Control/Data/Address Bus Multiplexing
- Serial Control/Data/Address Bus Multiplexing

The Parallel Access Multiplexing approach is the most recommended DFT methodology for the embedded FlashIP blocks to reduce the test time and the effort for the test program development.

If the number of I/O pads or pins available for the test of FlashIP blocks is a bit of concern, users can choose the Data/Address Bus Multiplexing approach to incorporate the data and address buses into a single Data/Address bus to reduce the test pads or pins needed for the test of FlashIP blocks.

Design for Testability (continued)

The Control/Data/Address Bus Multiplexing approach is applicable for low pin count applications. Furthermore, the Serial Control/Data/Address Bus Multiplexing approach is proposed specially for those very low pin count applications.

However, users may implement other ad-hoc DFT methods for testing the FlashIP blocks, if none of the four approaches is applicable to the DFT strategy of user application or product chip, for example, BIST (Built-In Self Test)

However, it is necessary to confirm with TSMC if the ad-hoc test methodology is feasible for the test program development and the testability of the FlashIP blocks. Otherwise, the embedded FlashIP blocks may not be testable for production test flow. In this case, the quality and reliability of FlashIP blocks can not be verified for volume production.

Design for Testability (continued)

◆ Parallel Access Multiplexing

- The Parallel Access Multiplexing approach is the most direct and simple way for the DFT of FlashIP blocks. This DFT approach is most recommended.

- While enabling the FlashIP test mode, the FlashIP blocks should be all made transparent to user I/O pads or I/O pins. All the interface pins of the FlashIP blocks are now multiplexed to the user I/O pins or pads.

- For the Parallel Access Multiplexing, the following signals must be multiplexed to user I/O pads: the XADR[i-1:0], YADR[j-1:0], DIN/DOUT[k-1:0], XE, YE, SE, ERASE, MAS1, PROG, NVSTR, IFREN, TMR, VPP, and TM[1:0], excluding the FlashIP-EN (FlashIP Test Enable).

The FlashIP-EN is not necessary to be a stand-alone test pad or test pin. It could be generated or enabled through other means. For example, the programming of user pins during the power-on reset, the trigger of test bits from other test-related registers through a sequence of settings, or the unused combination of user pins.

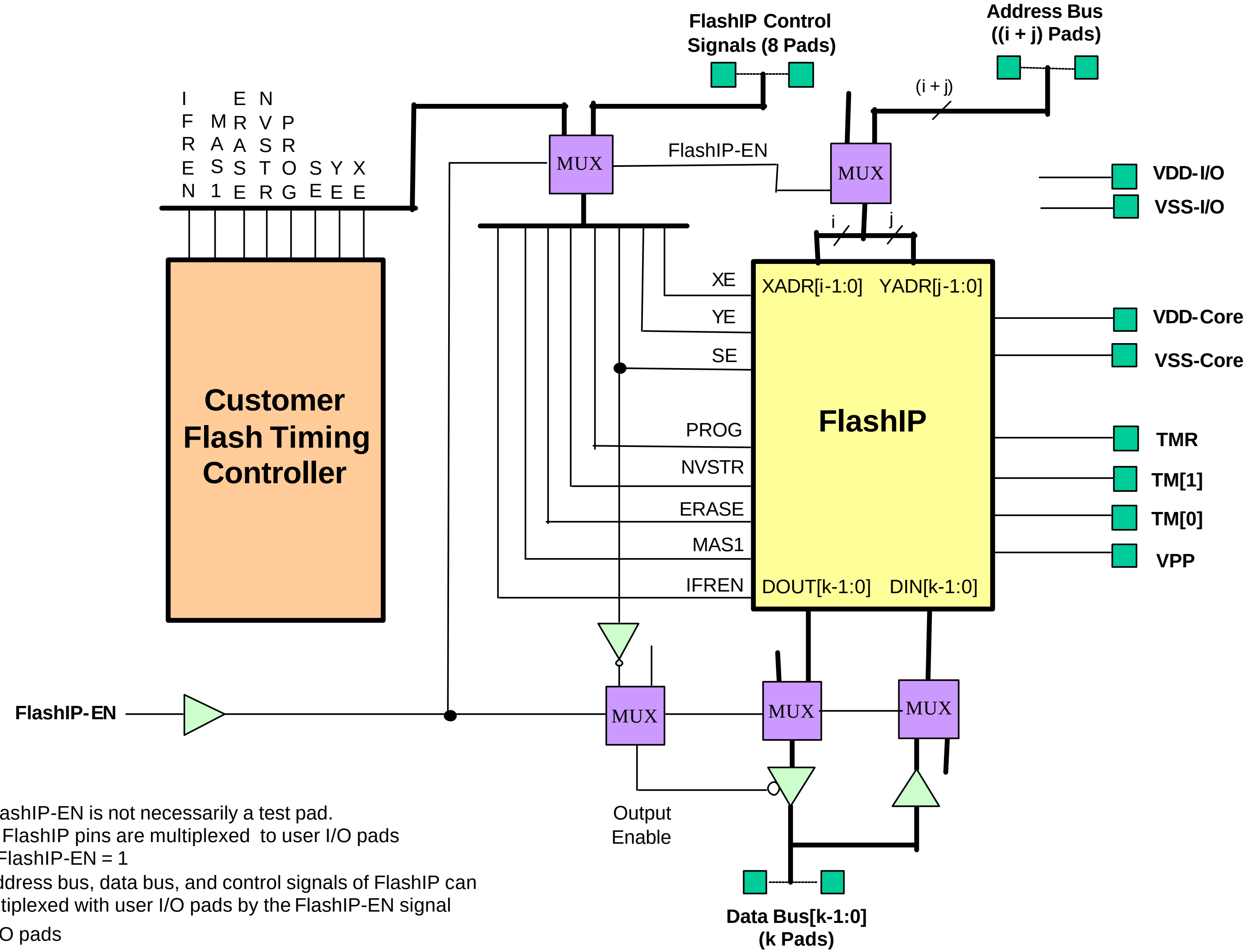
Design for Testability (continued)

- The number of test pads needed for the Parallel Access Multiplexing is ((i + j + k + 8 + 4) + 4), excluding the FlashIP-EN. The i, j and k pads are for the XADR[i-1], YADR[j-1], and DIN/DOUT[k-1] buses. The 8 pads are for the eight FlashIP control signals, and the 4 pads are for the TMR, VPP and TM[1:0]. The last 4 pads are for the power and ground pads of core circuit and I/O post-driver.

- The number of package pins needed for the Parallel Access Multiplexing is ((i + j + k + 8) + 4), excluding the TMR, VPP, TM[1:0] and FlashIP-EN. The i, j and k pins are for the XADR[i-1], YADR[j-1], and DIN/DOUT[k-1] buses The 8 pins are for the eight FlashIP control signals. The last 4 pins are for the power and ground pins of core circuit and I/O post driver.

If the capability of failure analysis and debugging for the embedded FlashIP block are planned to be kept for package parts, 4 extra pins should be assigned for the TMR, VPP and TM[1:0] pads.

Parallel Access Multiplexing



Notes:

1. The FlashIP-EN is not necessarily a test pad.
2. All the FlashIP pins are multiplexed to user I/O pads when FlashIP-EN = 1
3. The address bus, data bus, and control signals of FlashIP can be multiplexed with user I/O pads by the FlashIP-EN signal
4. ▢ : I/O pads

Design for Testability (continued)

◆ Data/Address Bus Multiplexing

- The XADR, YADR and DIN buses are latched or registered from the same Data/Address bus with the three latch enables of the XALE, YALE and DLE. The three enables are X-address Latch Enable, Y-address Latch Enable, and Data Latch Enable.

- The number of pads needed for the XADR[i-1], YADR[j-1], DIN[k-1] /DOUT[k-1] buses is now reduced to (max(i, j, k) + 3), the maximum width among the DIN/DOUT, XADR or YADR buses plus the three pads of the XALE, YALE and DLE.

- For the Data/Address Bus Multiplexing approach, the following signals must be multiplexed to user I/O pads: the Data/Address bus, XALE, YALE, DLE, XE, YE, SE, MAS1, PROG, ERASE, NVSTR, IFREN, TMR, VPP and TM[1:0], excluding the FlashIP-EN (FlashIP Test Enable).

The FlashIP-EN is not necessary to be a stand-alone test pad or test pin. It could be generated or enabled through other means. For example, the programming of user pins during the power-on reset, the trigger of test bits from other test-related

Design for Testability (continued)

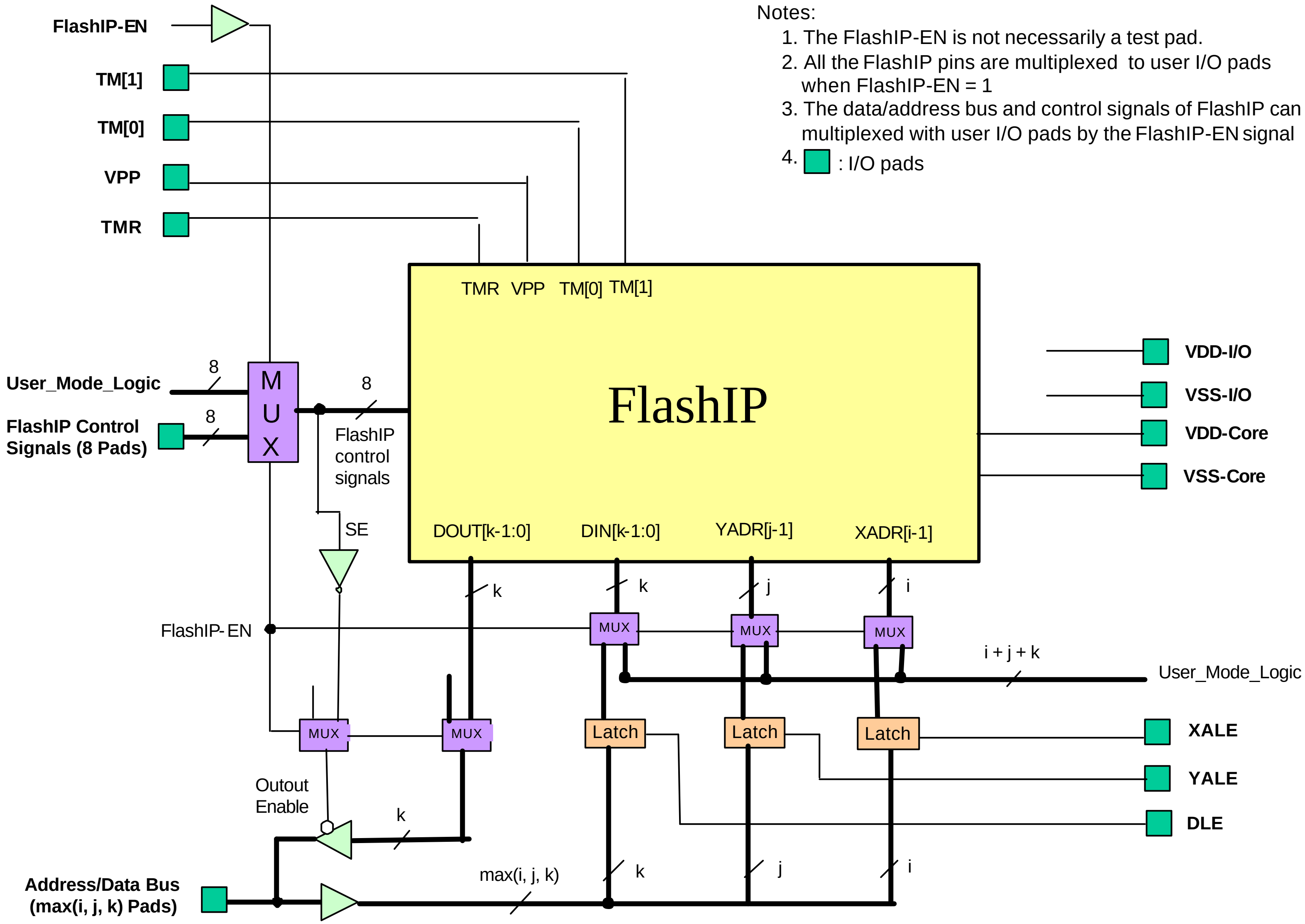
registers through a sequence of settings, or the unused combination of user pins.

- The number of test pads needed for the Data/Address Bus Multiplexing is ((max(i , j, k) + 3 + 8 + 4) + 4), excluding the FlashIP-EN. The max(i, j, k) pads are for the multiplexed Data/Address bus. The 3 pads are for the XALE, YALE and DLE, the 8 pads are for the eight FlashIP control signals, and the 4 pads are for the TMR, VPP and TM[1:0]. The last 4 pads are for the power and ground pads of core circuit and I/O post-driver.

- The number of package pins needed for the Data/Address Bus Multiplexing is ((max(i , j, k) + 3 + 8) + 4), excluding the TMR, VPP, TM[1:0] and FlashIP-EN. The max(i , j, k) pins are for the multiplexed Data/Address bus. The 3 pins are for the XALE, YALE and DLE, and the 8 pins are for the eight FlashIP control signals. The last 4 pins are for the power and ground pins of core circuit and I/O post driver.

If the capability of failure analysis and debugging for the embedded FlashIP block are planned to be kept for package parts, 4 extra pins should be assigned for the TMR, VPP and TM[1:0] pads.

Data/Address Bus Multiplexing



Notes:

1. The FlashIP-EN is not necessarily a test pad.
2. All the FlashIP pins are multiplexed to user I/O pads when FlashIP-EN = 1
3. The data/address bus and control signals of FlashIP can be multiplexed with user I/O pads by the FlashIP-EN signal
4. : I/O pads

Design for Testability (continued)

◆ Control/Data/Address Bus Multiplexing

● The Control/Address/Data Bus Multiplexing approach is applicable for low pin count applications. All the control signals, data bus, and address bus are multiplexed in or out with an common 8-bit Control/Data/Address bus in different time slots.

● With the three select signals, SEL[2:0], eight data modes on the Control/Address/Data Bus can be defined for the high-byte address, low-byte address, high-byte data-in, low-byte data-in, high-byte data-out, low-byte data-out, 8-bit control signals and 8-bit extra signals.

● For the Control/Data/Address Bus Multiplexing approach, the following signals must be multiplexed to user I/O pads: the 8-bit Control/Address/Data bus, SEL[2:0], CLK, VPP, and TM[1:0], excluding the FlashIP-EN (FlashIP Test Enable). The TMR signal could be generated from the latched data for extra signals in one of the defined data modes.

The FlashIP-EN is not necessary to be a stand-alone test pad or test pin. It could be generated or enabled through other means. For example, the programming of

Design for Testability (continued)

user pins during the power-on reset, the trigger of test bits from other test-related registers through a sequence of settings, or the unused combination of user pins.

- The number of test pads needed for the Control/Data/Address Bus Multiplexing is ((8 + 3 + 1 + 3) + 4), excluding the FlashIP-EN. The 8 pads are for the 8-bit Control/Data/Address bus, the 3 pads are for the three select controls, SEL[2:0], the 1 pad is for the clock signal, CLK, and the 3 pads are for the VPP and TM[1:0]. The last 4 pads are for the power and ground pads of core circuit and I/O post driver.

- The number of package pins needed for the Control/Data/Address Bus Multiplexing is ((8 + 3 + 1) + 4), excluding the VPP, TM[1:0] and FlashIP-EN. The 8 pins are for the 8-bit Control/Data/Address bus, the 3 pins are for the three select controls, SEL[2:0], and the 1 pin is for the clock signal, CLK. The last 4 pins are for the power and ground pins of core circuit and I/O post driver.

If the capability of failure analysis and debugging for the embedded FlashIP block are planned to be kept for package parts, 3 extra pins should be assigned for the three analog pads of the VPP and TM[1:0].

DFT Structure Table for Control/Address/Data Bus Multiplexing

SEL[2:0]	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Remark
000	A15	A14	A13	A12	A11	A10	A9	A8	High Byte Addr
001	A7	A6	A5	A4	A3	A2	A1	A0	Low Byte Addr
010	DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	High Byte Din
011	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Low Byte Din
100	DO15	DO14	DO13	DO12	DO11	DO10	DO9	DO8	High Byte Dout
101	DO7	DO6	DO5	DO4	DO3	DO2	DO1	DO0	Low Byte Dout
110	XE	YE	SE	IFREN	ERASE	PROG	MAS1	NVSTR	Control Signals
111	TMR	X	X	X	X	X	X	X	Extra Signals

(a) 8 Control/Address/Data pads : Bit 7, Bit 6, Bit 5, Bit 4, Bit 3, Bit 2 Bit 1, Bit 0

(b) 3 select pads : SEL[2], SEL[1], SEL[0] for mode selection

(c) 1 clock pad : CLK

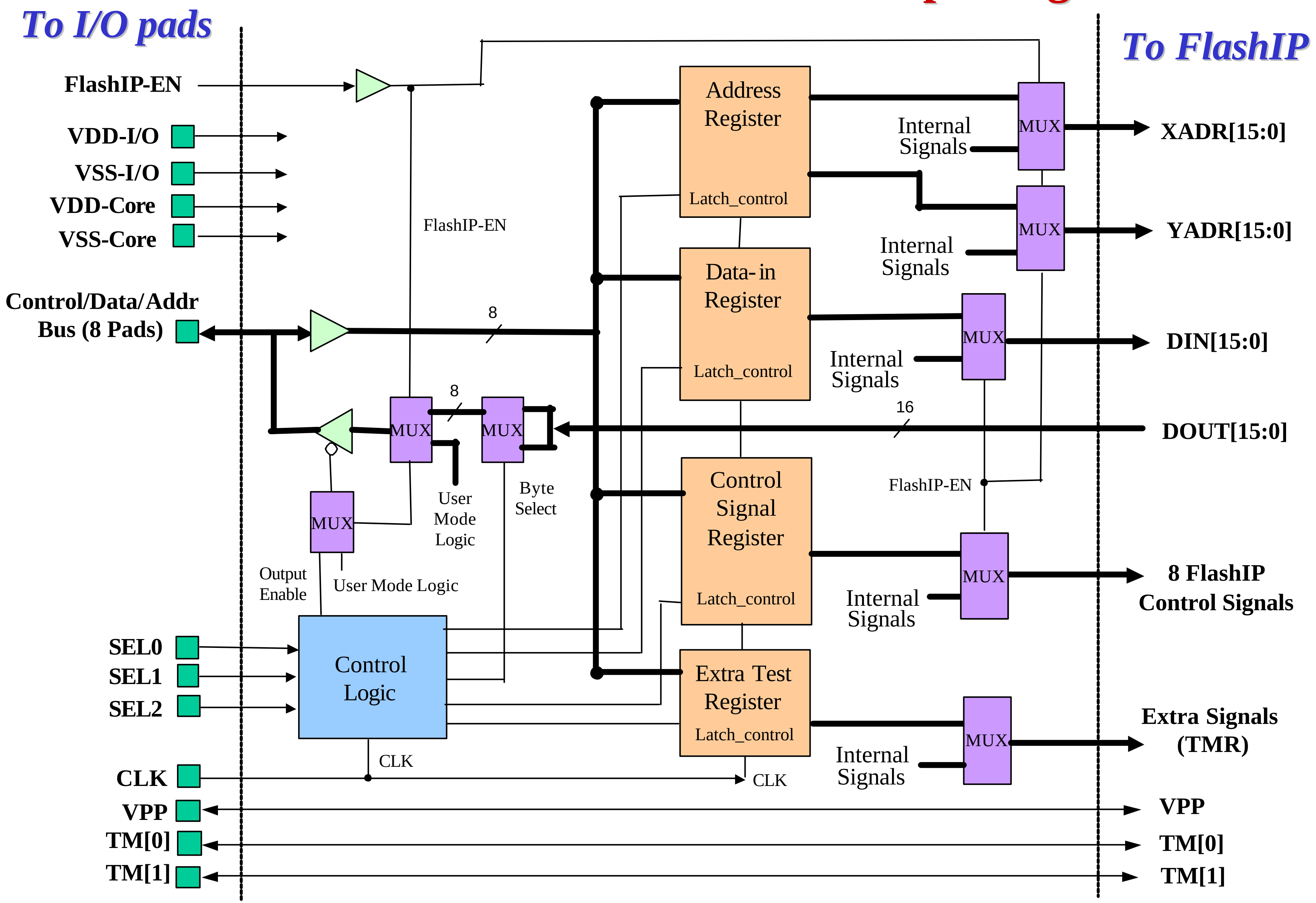
(d) 3 FlashIP specific test pads : VPP, TM[1:0]

(e) 4 power and ground pads : VDD-Core, VSS-Core, VDD-I/O and VSS-I/O

(f) Total test pads needed : ((8 + 3 + 1 + 3) + 4) pads, excluding the FlashIP-EN

(g) Total test pins needed : ((8 + 3 + 1) + 4) pins, excluding the VPP, TM[1:0], and FlashIP-EN

Control/Data/Address Bus Multiplexing



Design for Testability (continued)

◆ Serial Control/Data/Address Bus Multiplexing

● The Serial Control/Address/Data Multiplexing approach is suitable for very low pin count applications. This serial synchronous protocol reduces the interface pins needed for the test of the embedded FlashIP blocks to the three pins of the DI, DO and CLK.

● The DI, DO and CLK pins are used for read operation, and the DI and CLK pins are used for program operation, excluding the FlashIP-EN (FlashIP Test Enable).

● Twelve bits per data packet are sent for this approach. Bit 1 signalize the read or write operation. Bits 2~4, SEL[2:0], are used for the selection of data modes. Bits 5-12 are the data bits to be input from the DI pin or to be output to the DO pin.

● With the three select bits, SEL[2:0], eight data modes on the Control/Address/Data Bus can be defined for the high-byte address, low-byte address, high-byte data-in, low-byte data-in, high-byte data-out, low-byte data-out, 8-bit control signals and 8-bit extra signals.

Design for Testability (continued)

- For the Serial Control/Data/Address Bus Multiplexing approach, the following signals must be multiplexed to user I/O pads: DI, DO, CLK, VPP, and TM[1:0], excluding the FlashIP-EN (FlashIP Test Enable). The TMR signal could be generated from the latched data for extra signals in one of the defined serial data modes.

The FlashIP-EN is not necessary to be a stand-alone test pad or test pin. It could be generated or enabled through other means. For example, the programming of user pins during the power-on reset, the trigger of test bits from other test-related registers through a sequence of settings, or the unused combination of user pins.

- The number of test pads needed for the Serial Control/Data/Address Bus Multiplexing is ((3 + 3) + 4), excluding the FlashIP-EN. The first 3 pads are for the DI, DO, CLK, and the other 3 pads are for the VPP and TM[1:0]. The last 4 pads are for the power and ground pads of core circuit and I/O post driver.

- The number of package pins needed for the Serial Control/Data/Address Bus Multiplexing is ((3) + 4), excluding the VPP, TM[1:0] and FlashIP-EN. The 3 pins are for DI, DO and CLK. The last 4 pins are for the power and ground pins of core circuit and I/O post driver.

Design for Testability (continued)

If the capability of failure analysis and debugging for the embedded FlashIP block are to be kept for package parts, 3 extra pins should be assigned for the three analog pads of the VPP and TM[1:0].

- The DI input pin and DO output pin can be further combined to a single bi-directional pin to reduce the interface pins needed for the protocol to the two pins, the DI/DO and CLK pins.

- For a bi-directional data pin, an extra tri-state cycle should be inserted right before the pin changes its data direction either from input to output or from output to input, to avoid the potential data contention between the data input and data output.

- This action also increase the data packet of a read operation to 14 bits, taking 14 clock cycles. Please refer to the waveform of read cycle in the following for more details.

- The output enable of the bi-directional data pin is recommended to be activated for 8 clock cycles and inactivated automatically by internal control circuit, like state machine, to avoid the needless toggle on the output enable.

DFT Structure Table for Serial Control/Address/Data Bus Multiplexing

R/W	SEL[2:0]	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Remark
0/1	000	A15	A14	A13	A12	A11	A10	A9	A8	High Byte Addr
0/1	001	A7	A6	A5	A4	A3	A2	A1	A0	Low Byte Addr
0/1	010	DI15	DI14	DI13	DI12	DI11	DI10	DI9	DI8	High Byte Din
0/1	011	DI7	DI6	DI5	DI4	DI3	DI2	DI1	DI0	Low Byte Din
0/1	100	DO15	DO14	DO13	DO12	DO11	DO10	DO9	DO8	High Byte Dout
0/1	101	DO7	DO6	DO5	DO4	DO3	DO2	DO1	DO0	Low Byte Dout
0/1	110	XE	YE	SE	IFREN	ERASE	PROG	MAS1	NVSTR	Control Signals
0/1	111	TMR	X	X	X	X	X	X	X	Extra Signals

(a) Twelve bits per data packet are sent. Bit 1 signalizes read or write operation. Bits 2~4, SEL[2:0], are used for the selection of data modes. Bits 5-12 are the data bits for the DI or DO data bus.

(b) 3 pads: DI, DO and CLK

(c) 3 FlashIP specific test pads : VPP, TM[1:0]

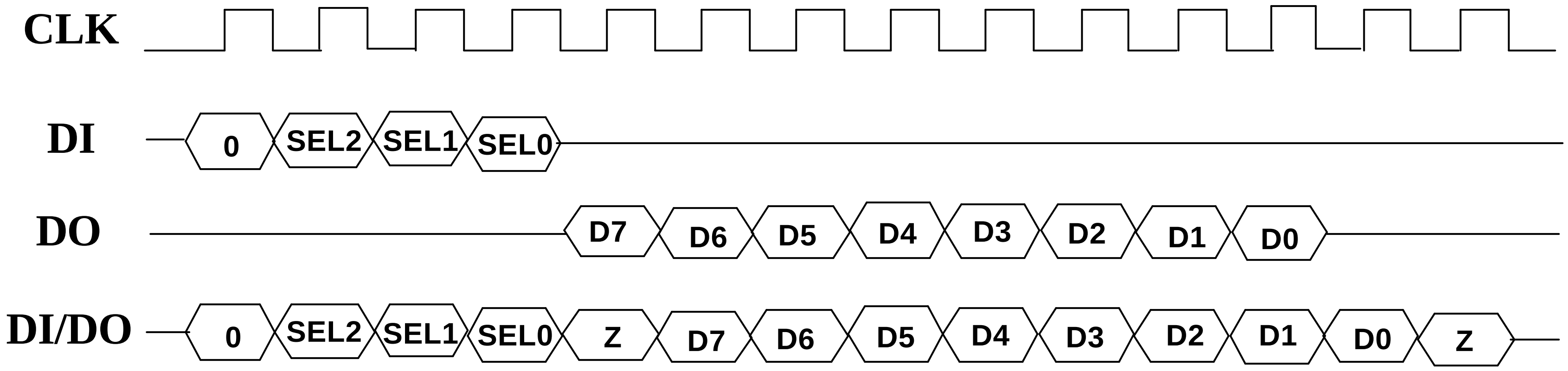
(d) 4 power and ground pads : VDD-Core, VSS-Core, VDD-I/O and VSS-I/O

(e) Total test pads needed : ((3 + 3) + 4) pads, excluding the FlashIP-EN

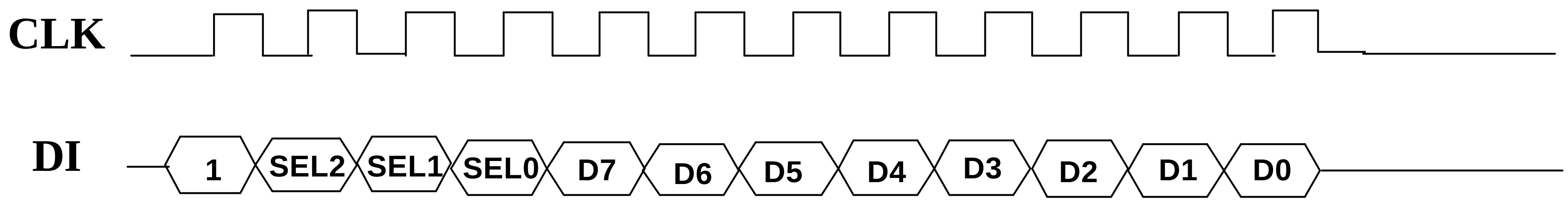
(f) Total test pins needed : ((3) + 4) pins, excluding the VPP, TM[1:0], and FlashIP-EN

Read/Write Waveform for Serial Control/Address/Data Bus Multiplexing

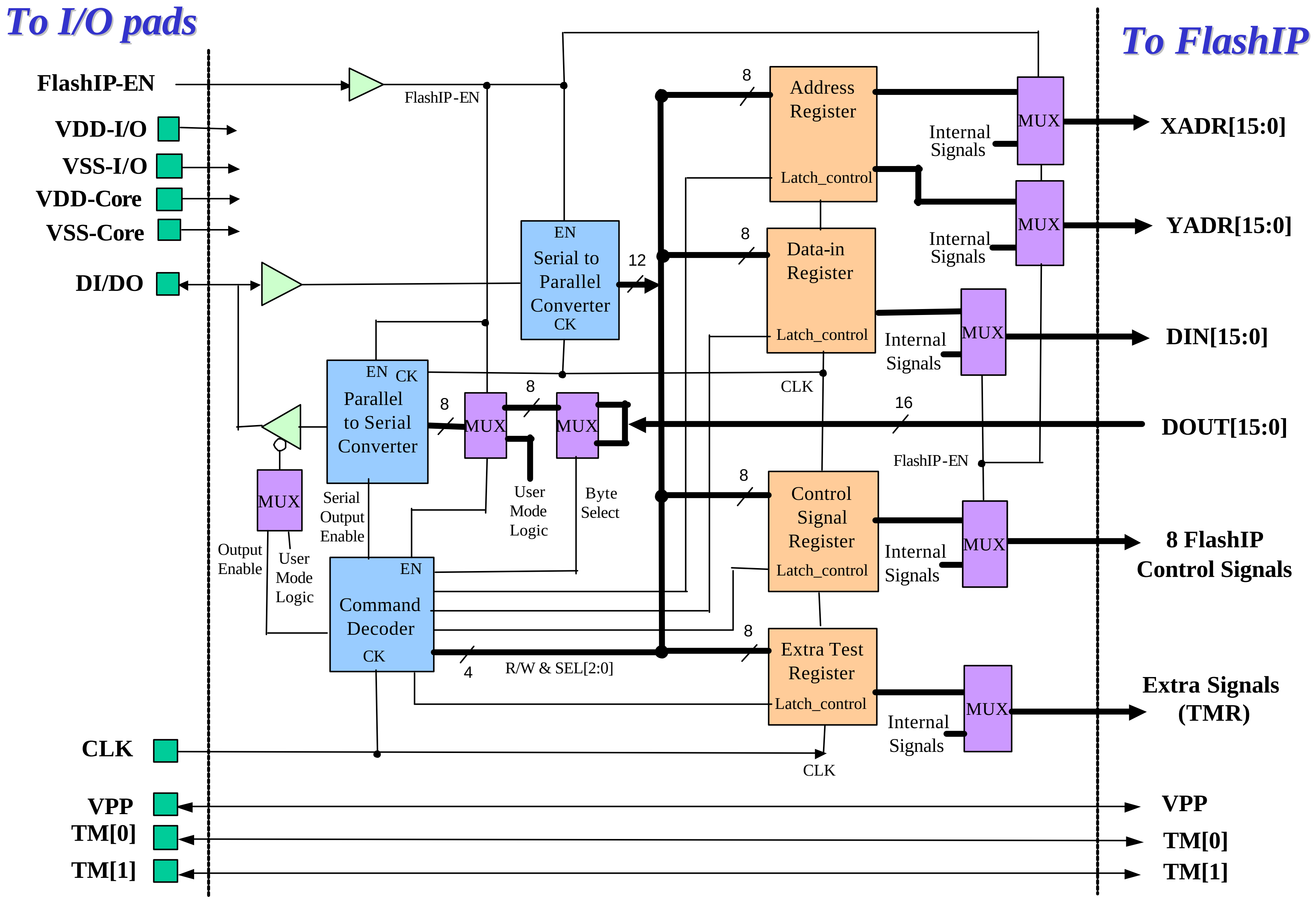
Read Cycle



Write Cycle



Serial Control/Data/Address Bus Multiplexing



FlashIP™ Design Kits

FlashIP Deliverables

- Behavior Simulation Model
  - Vital/VHDL
  - Verilog
- Synthesis
  - Synopsys Lib/DB
- P&R tool
  - LEF View of Cadence for First Encounter/Silicon Ensemble
  - FRAM View of Synopsys for Astro/Apollo
  - GDSII phantom (black box) for other P&R tools.

HV Pad Deliverables

- Behavior Simulation Model
  - Vital/VHDL
  - Verilog
- Synthesis
  - Synopsys Lib/DB
- GDSII layout

FlashIP™ Request Form and TSA

The filled FlashIP Request Form and signed TSA (Technology Support Agreement) for EmbFlash are required for the application of embedding the TSMC FlashIP blocks. Please contact the TSMC support staff or Account Manager for the blank form and agreement.

● FlashIP Request Form

This request form is intended to evaluate if the FlashIP specification is able to match the application or design requirements of customer products. Be noted that the form is served for evaluation purpose only. That is, it is not a commitment to customer requirements.

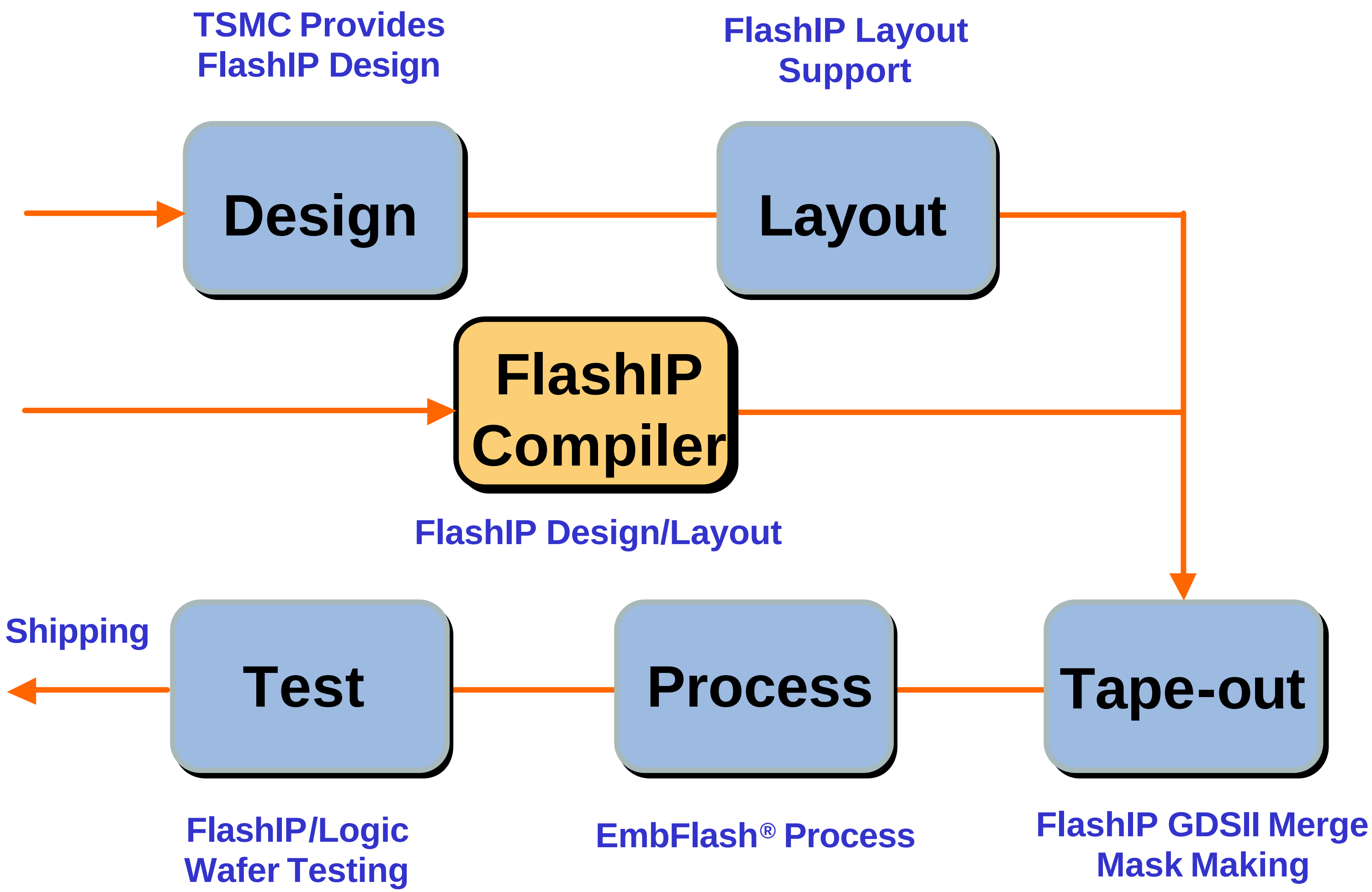
● Technology Support Agreement

The signed TSA is the prerequisite for the release of the front-end design kits and phantom design kits of FlashIP blocks. It also sub-license the right to the use of TSMC FlashIP to customers.

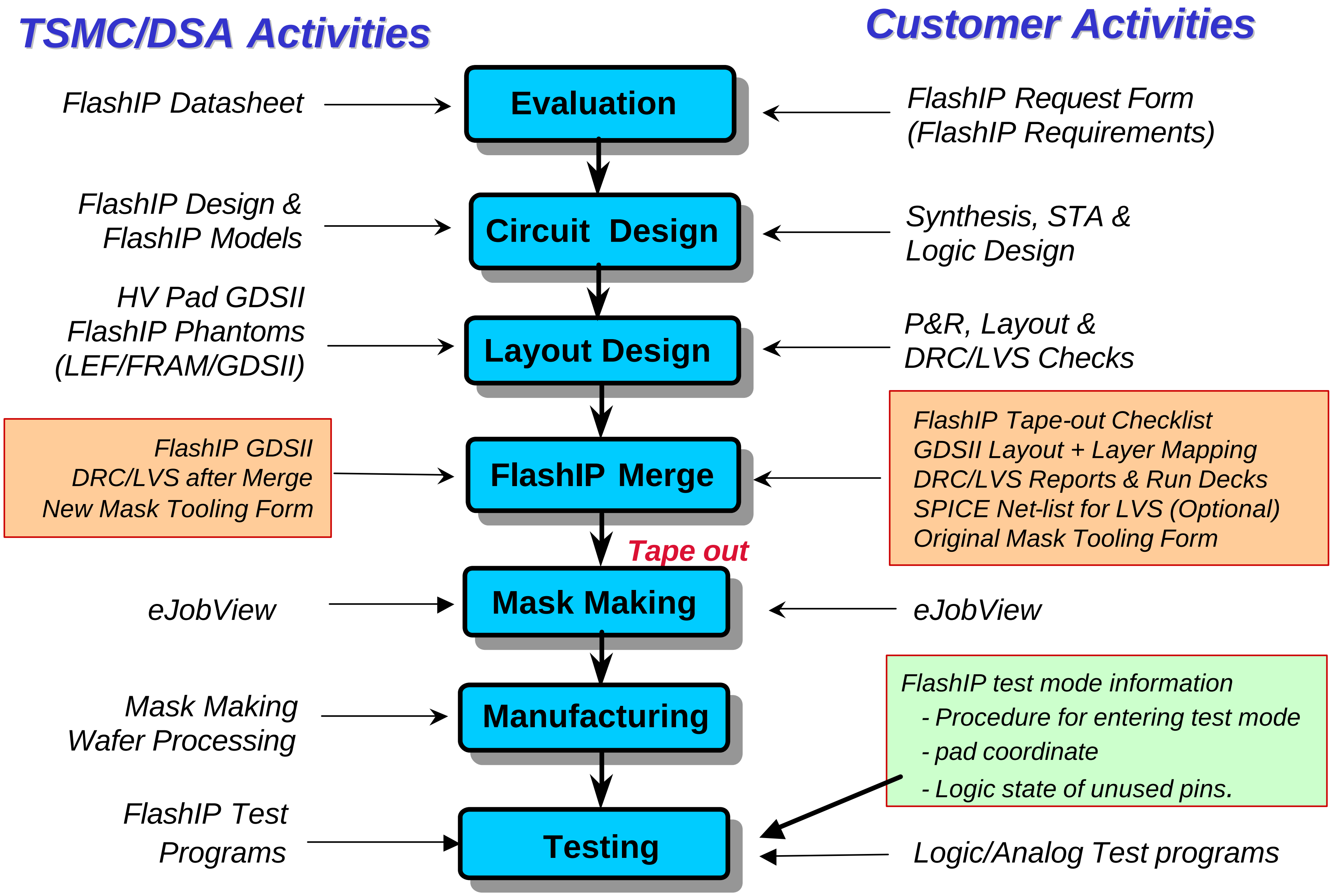
FlashIP™ Data Available

- **Information Available on TSMC-ONLINE:**
  - FlashIP Spec
  - HV pad Spec
  - FlashIP application Note
  - Spice model /PCM spec
  - EmbFlash/Mixed-Mode/Logic Design Rules
  - EmbFlash DRC/LVS run decks for Calibre of Mentor Graphics
  - and Hercules of Synopsys
  - Tape-out Checklist
- **Other Library Kits**
  - TSMC In-house Analog I/O Library (for TM[1:0] analog pads)
  - TSMC In-house Standard I/O Library
  - TSMC In-house Specialty I/O Library
  - TSMC In-house Mixed Signal/RF Library

FlashIP™ Service/Support Flow

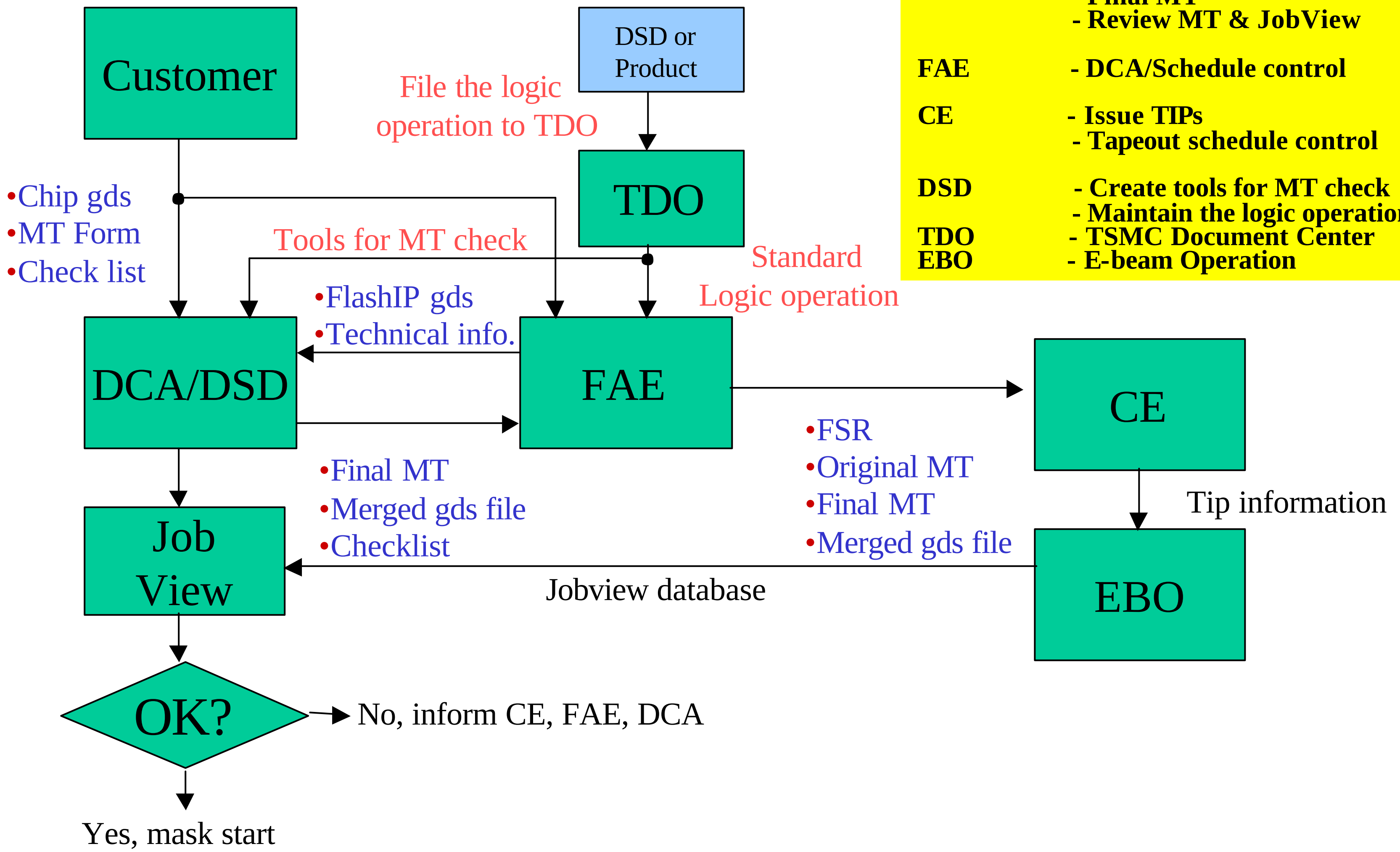


FlashIP™ Service/Support Model



FlashIP™ GDSII Merge Flow

Party	Function
DCA/DSD	- IP merge - Final MT - Review MT & JobView
FAE	- DCA/Schedule control
CE	- Issue TIPs - Tapeout schedule control
DSD	- Create tools for MT check - Maintain the logic operation
TDO	- TSMC Document Center
EBO	- E-beam Operation



Data Prep for FlashIP™ Merge

- Filled FlashIP Merge checklist
- Chip GDSII database compliant with TSMC standard layer definition
- Layer mapping/definition file for the released GDSII except for the FlashIP
- Clear DRC and DRC verification reports (DRC/LVS/ANT/Metal Density)
- The verification run deck version (DRC/LVS/ANT/Metal Density)
- Optional chip SPICE netlist for LVS check after FlashIP merge
- Filled MT information, including the following information:
  - Process rule, Library name and Top cell name
  - Cell size and Cell window coordinates
  - Mask layer number, GDS layer number
  - Customer logic operations, except those for FlashIP

Data Prep for FlashIP™ Tapeout

- **MT information**
  - Clear layer mapping/definition
  - Clear logic operations for the layout except for FlashIP
- **Flash test document**
  - Top level block diagram for FlashIP test interface
  - Procedure for entering into the flashIP test mode
  - Bonding diagram for probe card manufacturing
  - Pin description and assignment for FlashIP test mode
  - Logic test patterns, if logic test is required at wafer sort-2
  - Additional timing waveforms, if exist
- **LVS/DRC after IP merge**
  - DRC check is default
  - LVS check is optional
    - Provide the SPICE netlist to TSMC
    - Provide the customized LVS run deck to TSMC, if exists
  - Generate new MT Form for mask making

FlashIP™ CP/FT Tape-out Review

The following information are needed for the CP/FT testability reviewed by TSMC, when requested, before the product tape-out with the embedded FlashIP blocks. In particular, it is strongly recommended to be performed for the first tape-out of the newcomers at integrating the embedding the FlashIP blocks.

- **Top level block diagram for the FlashIP test interface**

  The schematic diagram includes the FlashIP block names, FlashIP pin names, user I/O pad names, user I/O pin names, and other glue logic for the FlashIP test mode.

- **Procedure to enter the FlashIP test mode**

  For example, RESET = VDD, CNTL0 = LOW, CNTL1 = LOW

  Refer to the block diagram examples of the four DFT approaches introduced in the Design for Testability session of this application note.

- **Bonding diagram for probe card manufacturing**

FlashIP™ CP/FT Tape-out (continued)

- **Pin description and assignment for the FlashIP test mode. For example,**

PIN #	USER MODE	FLASH TEST MODE	Comment
1	RESET	HIGH	Must be HIGH to enter FlashIP test mode
2	CNTL0	LOW	Must be LOW to enter FlashIP test mode
3	CNTL1	LOW	Must be LOW to enter FlashIP test mode
4	ADDR0	XADR[0]	
5	ADDR1	XADR[1]	
6	LD0	DIN[0]/DOUT[0]	Equal to DIN[0] when SE = 0
7	LD1	DIN[1]/DOUT[1]	Equal to DIN[0] when SE = 0
• • •	• • •	• • •	• • •

- **Logic test patterns, if additional logic test is required at wafer sort-2**
- **Additional timing waveforms, if exist**

FlashIP™ CP/FT Reminder

The at-speed timing test is not included or supported in the TSMC standard wafer sort service.

If the at-speed test, especially for the read access, is required for user applications, it is recommended to be done on package parts at the final test stage with a checkerboard pattern and at-speed timing.

All the pins which require certain particular states during the setup or access of the embedded FlashIP blocks must be declared clearly. Those unspecified pins will be floated during the test of the the FlashIP blocks. Any missing or undeclared pin generally causes test failure. Sometimes a new probe card is needed to be remade, causing extra hardware cost. Most importantly, longer delay time for the completion of the FlashIP test will be suffered.

The default test channels are only wire-connected. Customers must clearly specify the existing pull-up, pull-down, or open-drain pins, and so on. Otherwise, the test failure will be hardly debugged by test engineers.

Revision History

Rev	Date	From	Description
1.3	05/16/2003	YA Liang	Original
1.4	07/29/2003	YA Liang	1. Restore the program time from 20us to 20us-40us on p. 8. 2. Correct the timing waveform of Word (Byte) Read on p. 9. 3. Add the "Multiple FlashIP Considerations" session on pp. 33-35. 4. Other wordings.